



Eosinophil extracellular traps drive asthma progression through neuro-immune signals

Yiwen Lu^{1,2,9}, Yijiao Huang^{1,3,9}, Jiang Li^{1,2,9}, Jingying Huang^{1,2,9}, Lizhi Zhang^{1,3,9}, Jingwei Feng^{1,2}, Jiaqian Li^{1,2}, Qidong Xia^{1,2}, Qiyi Zhao^{4,5,6}, Linjie Huang^{1,3,7}, Shanping Jiang^{1,3,7}  and Shicheng Su^{1,2,4,8} 

Eosinophilic inflammation is a feature of allergic asthma. Despite mounting evidence showing that chromatin filaments released from neutrophils mediate various diseases, the understanding of extracellular DNA from eosinophils is limited. Here we show that eosinophil extracellular traps (EETs) in bronchoalveolar lavage fluid are associated with the severity of asthma in patients. Functionally, we find that EETs augment goblet-cell hyperplasia, mucus production, infiltration of inflammatory cells and expressions of type 2 cytokines in experimental non-infection-related asthma using both pharmaceutical and genetic approaches. Multiple clinically relevant allergens trigger EET formation at least partially via thymic stromal lymphopoietin in vivo. Mechanically, EETs activate pulmonary neuroendocrine cells via the CCDC25-ILK-PKC α -CRT1 pathway, which is potentiated by eosinophil peroxidase. Subsequently, the pulmonary neuroendocrine cells amplify allergic immune responses via neuropeptides and neurotransmitters. Therapeutically, inhibition of CCDC25 alleviates allergic inflammation. Together, our findings demonstrate a previously unknown role of EETs in integrating immunological and neurological cues to drive asthma progression.

Asthma, which occurs in approximately 300 million people worldwide, is one of the most common chronic disorders and poses a major challenge for global health and healthcare costs^{1–4}. Eosinophil infiltration of lung tissues is one of the key pathological features of asthma¹. It has been reported that eosinophil infiltration of lung mucosa and interstitium is closely associated with the type 2 immune response and the severity of asthma⁴. Depletion of eosinophils alleviates the progression of asthma in mouse models⁵. More importantly, humanized antibodies targeting interleukin (IL)-5 (an eosinophil-stimulating cytokine) have been approved for adult and adolescent patients with eosinophilic asthma, highlighting the crucial role of eosinophils in human asthma^{6–8}. Therefore, dissecting how eosinophils drive airway inflammation holds great promise for improving asthma management.

Neutrophil extracellular traps (NETs) are extracellular chromatin DNA fibres impregnated with granule proteins. NETs are traditionally considered as a neutrophil weaponry to capture infectious microorganisms. However, the paradigm has been shifted as accumulating evidence demonstrates that NETs play a crucial role in various types of non-infectious diseases^{9–14}. Similar to neutrophils, eosinophils can generate web-like DNA traps—eosinophil extracellular traps (EETs). Despite NETs being at the forefront of biomedical research, the pathogenic functions of EETs have received little attention. We recently identified CCDC25, a transmembrane protein on the cytoplasmic membrane, as a specific sensor for NET-DNA on tumour cells that promotes distant metastasis¹⁵. Whether CCDC25 also plays a role in other non-tumoral diseases is unknown.

Recent studies show that neuro-immune interactions are fundamentally critical to maintain health. The corruption of neuronal and immune cell communication drives disease progression^{16–18}. For example, pulmonary neuroendocrine cells (PNECs) amplify allergic asthma inflammation via neuropeptides and neurotransmitters, including calcitonin gene-related peptide (CGRP)¹⁹. However, the mechanisms of PNEC activation during asthma are unclear. Here we investigate the role of EETs in asthma and its underlying mechanisms.

Results

EET levels correlate with asthma progression. To investigate the clinical significance of EETs in the progression of asthma, we obtained bronchoalveolar lavage fluid (BALF) from patients with asthma and healthy volunteers by fibre-optic bronchoscopy. Using electron and immunofluorescence microscopy, we observed EETs—web-like DNA structures co-localizing with eosinophil cationic protein (ECP)²⁰—in smears of the BALF sediments of most of the patients with asthma (42/46) but not the volunteers without asthma (Fig. 1a,b). Furthermore, we quantitated EETs as the area of decondensed thread-like DNA co-localizing with ECP as previously reported^{21,22} and found that the levels of EETs correlated closely with the severity of asthma, which was measured as the proportion of the forced expiratory volume in one second to forced vital capacity (FEV₁%), and the Global Initiative for Asthma (GINA) guideline-defined classification levels (Fig. 1c). These data indicated that EETs are associated with asthma progression.

¹Guangdong Provincial Key Laboratory of Malignant Tumor Epigenetics and Gene Regulation, Medical Research Center, Sun Yat-Sen Memorial Hospital, Sun Yat-Sen University, Guangzhou, China. ²Breast Tumor Center, Sun Yat-Sen Memorial Hospital, Sun Yat-Sen University, Guangzhou, China. ³Department of Pulmonary and Critical Care Medicine, Sun Yat-Sen Memorial Hospital, Sun Yat-Sen University, Guangzhou, China. ⁴Department of Infectious Diseases, The Third Affiliated Hospital, Sun Yat-Sen University, Guangzhou, China. ⁵Guangdong Provincial Key Laboratory of Liver Disease Research, The Third Affiliated Hospital, Sun Yat-Sen University, Guangzhou, China. ⁶Key Laboratory of Tropical Disease Control (Sun Yat-sen University), Ministry of Education, Guangzhou, China. ⁷Institute of Pulmonary Diseases, Sun Yat-Sen University, Guangzhou, China. ⁸Department of Immunology, Zhongshan School of Medicine, Sun Yat-Sen University, Guangzhou, China. ⁹These authors contributed equally: Yiwen Lu, Yijiao Huang, Jiang Li, Jingying Huang, Lizhi Zhang. ✉e-mail: jiangshp@mail.sysu.edu.cn; sushch@mail.sysu.edu.cn

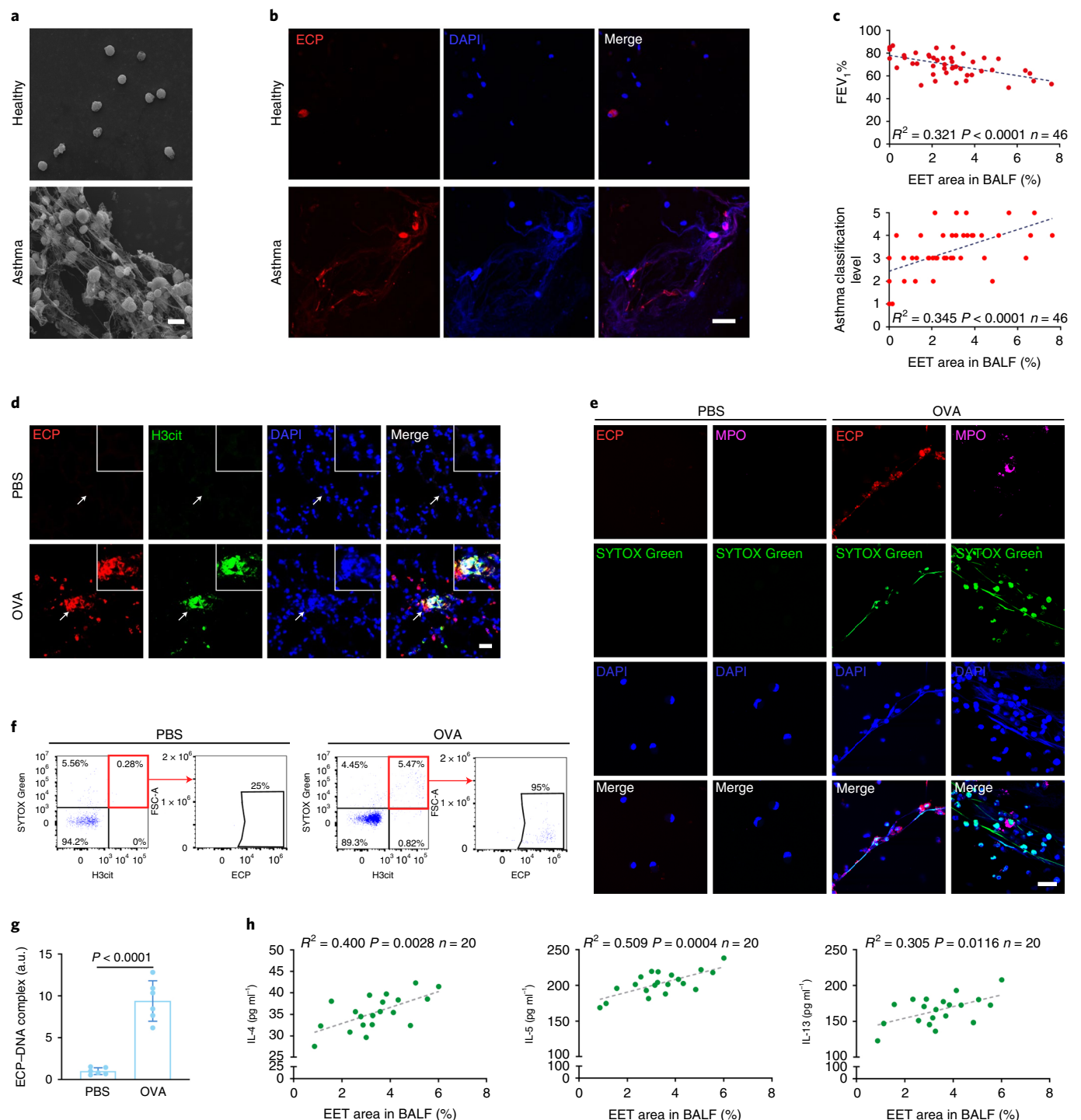


Fig. 1 | EET levels correlate with asthma progression. **a**, Representative scanning electron microscopy images of BALF sediments of healthy volunteers (top) and patients with asthma (bottom). Scale bar, 10 μ m. **b**, Representative immunofluorescence images of ECP and 4,6-diamidino-2-phenylindole (DAPI) staining of BALF smears of healthy volunteers (top) and patients with asthma (bottom). Scale bar, 50 μ m. **c**, Correlation between the EET area in immunofluorescence images of BALF smears and the FEV₁ values of patients with asthma (top) as well as the GINA guideline-defined classification levels (bottom). Pearson's (top) and Spearman's order (bottom) correlation analyses were performed. **d–g**, C57BL/6 mice were sensitized and challenged with PBS or OVA. **d**, Representative immunofluorescence images of ECP, H3cit and DAPI staining of the mouse lung sections. The arrows point to the area shown at a higher magnification in the insets. **e**, Representative immunofluorescence images of SYTOX Green, DAPI, ECP and MPO staining of BALF smears. **d, e**, Scale bars, 20 μ m. **f**, Representative flow cytometry analysis of SYTOX Green⁺H3cit⁺ECP⁺ cells in the BALF of mice challenged with PBS (left) or OVA (right). The percentage of cells in the respective quadrants or regions are indicated. **g**, The levels of EETs in the BALF were evaluated by ELISA. Data are the mean $\pm$ s.d.; $n = 6$ mice per group, representative of $n = 2$ independent experiments; compared by two-tailed unpaired Student's *t*-test; a.u., arbitrary units. **h**, Correlation between the EET area in immunofluorescence images of BALF smears and the concentration of the indicated type 2 cytokines, detected by ELISA, in the BALF of the OVA-treated mice. Data are pooled from $n = 3$ independent experiments with $n = 6$ –8 mice each. The square of the Pearson's correlation coefficient and *P* values are shown. R^2 , the square of the correlation coefficient.

To explore whether the presence of EETs in the airways of patients with asthma can be recapitulated in mouse models, we employed the well-established murine ovalbumin (OVA)-induced asthma model and found that fibrous structures of DNA co-localizing with ECP in the lung tissues of mice treated with OVA instead of PBS (Fig. 1d and Extended Data Fig. 1a). Moreover, we evaluated EETs in BALF samples using three different approaches—that is, immunofluorescence imaging²³, flow cytometry²⁴ and enzyme-linked immunosorbent assay (ELISA)¹⁵. Immunofluorescence staining for SYTOX Green, a plasma membrane-impermeable DNA dye²⁵, demonstrated fibrous structures of extracellular DNA co-localizing with ECP in the BALF sediments of the mice treated with OVA instead of PBS (Fig. 1e and Extended Data Fig. 1b). These imaging data were corroborated by flow cytometric analysis (Fig. 1f and Extended Data Fig. 1c) as well as an ELISA assay using anti-ECP as the capture antibody and anti-DNA as the detection antibody (Fig. 1g).

It has been reported that NETs play a critical role in infection-related asthma induced by lipopolysaccharides¹¹ and rhinovirus²⁶. By comparison, the number of eosinophils that infiltrate the lungs is much higher than that of neutrophils in mice with OVA-induced asthma^{5,27,28}. We observed that $57.09 \pm 10.23\%$ of the SYTOX Green staining signals co-located with ECP. In contrast, only $5.27 \pm 3.32\%$ of the SYTOX signals co-located with myeloperoxidase (MPO; Fig. 1e and Extended Data Fig. 1d).

In addition, the type 2 immune response is a key feature of asthma progression in this model²⁹. We found that the levels of EETs in the BALF correlated positively with the levels of type 2 cytokines—that is, IL-4, IL-5 and IL-13 (Fig. 1h). Together, these results suggest that EETs are closely associated with the severity of asthma in both humans and mice.

Generation and components of EETs. Given that OVA can drive the formation of EETs, we investigated whether clinically relevant allergens have similar effects. We therefore challenged mice with different allergens for asthma, including house dust mite (HDM), cockroach and pollen³⁰. Phorbol 12-myristate 13-acetate (PMA) was applied as a positive control. Using immunostaining for SYTOX Green and ECP in the BALF samples, we observed that these clinically relevant allergens could also induce the generation of EETs in vivo (Fig. 2a,b).

Eosinophils are typically activated by cytokines that are induced by allergens in vivo, including IL-4, IL-5, IL-13, IL-33, thymic stromal lymphopoietin (TSLP) and IL-25 (refs. ^{29,31,32}). Thus, we treated OVA-challenged mice with neutralizing antibodies against these cytokines and found that blockade of TSLP, but not the other cytokines, significantly reduced OVA-induced EETs (Fig. 2c,d). In agreement with this, we observed similar effects in asthmatic mice challenged with other allergens (Fig. 2e). Together, these results demonstrate that multiple allergens initiate the formation of EETs at least partially through TSLP.

To obtain qualitative proteomics of EETs, we released DNA-binding proteins with DNase I (Fig. 2f, lane 6) from PMA-treated human eosinophils and evaluated the protein composition using liquid chromatography with tandem mass spectrometry (LC-MS/MS). As controls, DNase I-free media from washed PMA-stimulated eosinophils (lane 4) were also subjected to the LC-MS/MS analysis. A candidate was defined as an EET-binding protein when it met the identification criteria in at least two of three independent samples³³. We identified 32 proteins: five eosinophil granule, five nuclear, seven cytosolic, six cytoskeleton, six organelle and three membrane proteins (Fig. 2g, Extended Data Fig. 2a and Supplementary Table 1).

In addition, we quantified the relative abundance of mitochondrial and chromosomal DNA in the EETs using the ratio of mitochondrial 16S DNA to chromosomal 18S DNA. Our data showed that EET-DNA contained $8.72 \pm 1.99\%$ mitochondrial DNA,

suggesting that EET-DNA is predominantly derived from chromosomal DNA (Fig. 2h).

DNase I treatment ameliorates asthma in mice. We intraperitoneally sensitized mice with OVA on days 0 and 7, and challenged them with OVA intratracheally on days 14, 15 and 16 (Fig. 3a). The OVA-immunized mice displayed augmented goblet-cell hyperplasia, mucus overproduction (Fig. 3b,c), excessive infiltration of inflammatory cells—including macrophages, T cells, Th2 cells, B cells and eosinophils (Fig. 3d–f and Extended Data Fig. 3a)—and lung damage (Fig. 3g,h and Extended Data Fig. 3b). In agreement with previous reports^{5,27}, few neutrophils were observed in the lungs of the OVA-challenged mice (Fig. 3d,e). In addition, the levels of type 2 cytokines were substantially higher in the BALF of the mice receiving OVA (Fig. 3i). To investigate the effect of extracellular double-stranded DNA (dsDNA), we administered DNase I intratracheally 8 h before challenge with OVA on days 14, 15 and 16. These asthmatic inflammatory responses were ameliorated by DNase I treatment (Fig. 3b–i and Extended Data Fig. 3a,b). No protease contamination of the DNase I, which we used, was detected by the protease activity assay (Extended Data Fig. 3c).

In addition to extracellular traps of granulocytes, extracellular dsDNA may arise from the death of local lung cells in airway inflammation²⁶. To evaluate the extent of local lung-cell death, we performed cytokeratin and terminal deoxynucleotidyl transferase dUTP nick-end labelling (TUNEL) staining at both early (day 15) and late (day 17) time points. Our data showed that the levels of cytokeratin⁺TUNEL⁺ dead lung epithelial cells increased only slightly on day 17 (Extended Data Fig. 3d,e), whereas the EETs in the BALF were markedly elevated (Extended Data Fig. 3f). Notably, the number of eosinophils in the BALF of the mice treated with and without DNase I were comparable on day 15, whereas the mice treated with DNase I had significantly fewer eosinophils in BALF on day 17 compared with those without DNase I treatment (Extended Data Fig. 3g). Moreover, it has been shown that DNase can inhibit dendritic-cell recruitment in rhinovirus-induced asthma exacerbation²⁶. We observed that treatment with DNase I did not significantly alter the dendritic-cell recruitment on day 15, despite a significant reduction on day 17 (Extended Data Fig. 3h,i). Together, these data suggest that the dsDNA from EETs is the main source of extracellular dsDNA in the lungs at the early stage of asthma, which aggravates airway inflammation.

EETs amplify allergic asthma responses. The formation of NETs can be reduced by the inhibition of peptidylarginine deiminase 4 (PAD4), which modulates chromatin decondensation by catalysing histone hypercitrullination³⁴. Thus, we investigated whether PAD4 also regulates the formation of EETs. Western blotting and immunofluorescence analysis showed comparable levels of PAD4 in human neutrophils and eosinophils, which were much higher than those of other immune cells (Extended Data Fig. 4a,b). We treated eosinophils with BB-Cl-amidine, a haloacetamide-based warhead that mimics arginine guanidinium, to inhibit the activity of peptidylarginine deiminase (PAD)³⁵. Human eosinophils exposed to BB-Cl-amidine exhibited much lower levels of citrullinated histone H3 (H3cit; Fig. 4a), chromatin decondensation³⁶ (Fig. 4b and Extended Data Fig. 4c) and EET formation (Fig. 4c and Extended Data Fig. 4d) after PMA stimulation. To specifically investigate the role of PAD4 in EET generation, we immunized mice deficient in PAD4 (*PAD4*^{−/−}) with multiple allergens and found that the levels of EETs in the BALF of the *PAD4*^{−/−} mice were markedly decreased compared with the wild-type mice (Fig. 4d). Together, these data indicate that PAD4 is required for the generation of EETs.

Similar to the ex vivo experiments, we sensitized *PAD4*^{−/−} mice with OVA. Co-localization of extracellular H3cit staining and dsDNA in the ECP-positive areas was significantly less than the

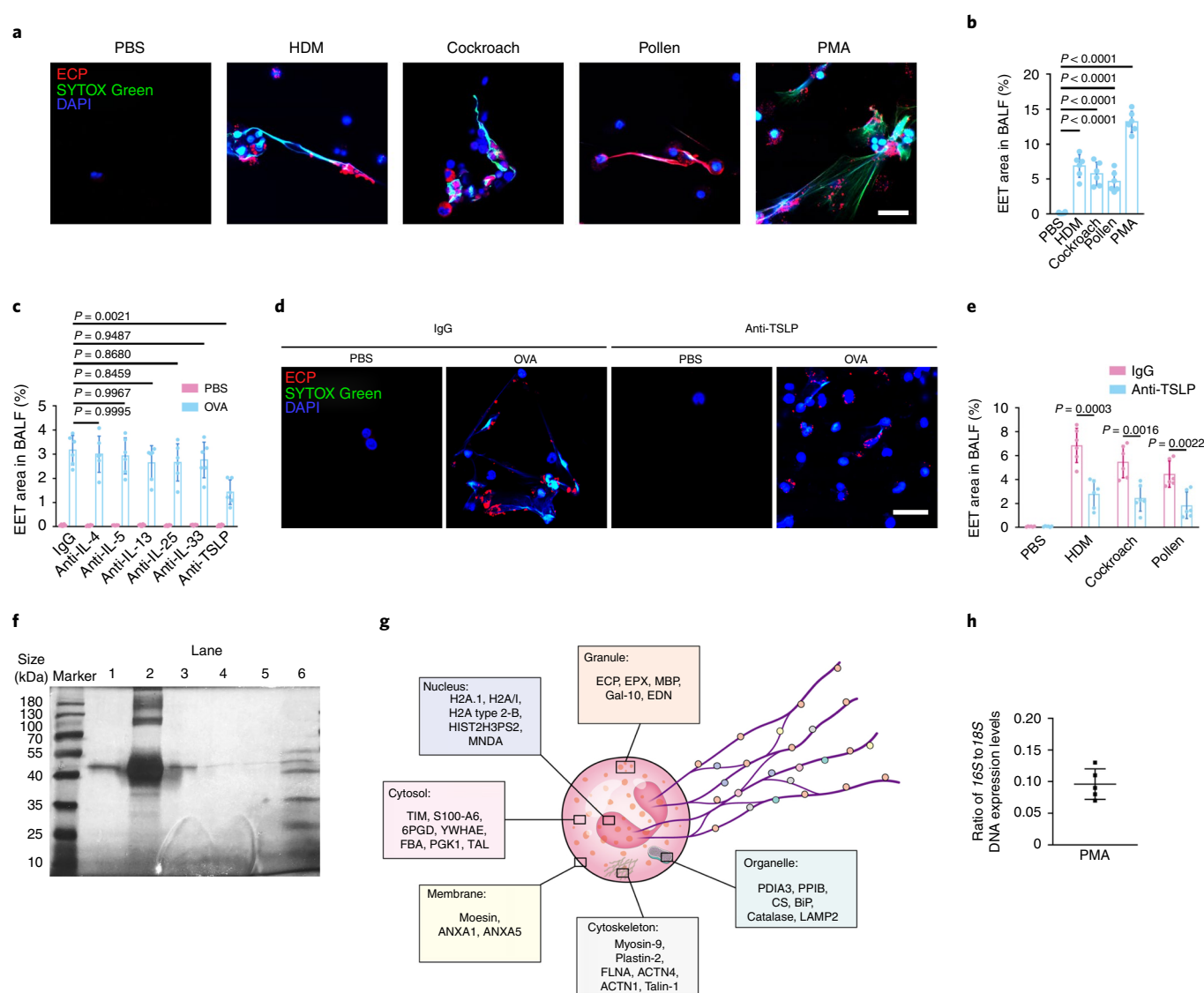


Fig. 2 | Generation and components of EETs. **a, b**, C57BL/6 mice were sensitized and challenged with HDM, cockroach and pollen. Mice receiving intratracheal PMA instillation served as positive controls. Representative immunofluorescence images of ECP, SYTOX Green and DAPI staining of BALF smears (**a**) and quantification of the EETs (**b**). **c**, C57BL/6 mice were treated with the indicated antibodies before being challenged with OVA. The ECP+SYTOX Green⁺ EET area in immunofluorescence images of BALF smears is shown. **d**, C57BL/6 mice were treated with IgG or a neutralizing antibody against TSLP before being challenged with OVA. Representative immunofluorescence images of ECP, SYTOX Green and DAPI staining of BALF smears are shown. **a, d**, Scale bars, 20 μ m. **e**, C57BL/6 mice immunized with HDM, cockroach or pollen were treated with IgG or a TSLP-neutralizing antibody. The area of ECP+SYTOX Green⁺ EETs in BALF smears was determined. **f–h**, Eosinophils isolated from peripheral blood mononuclear cells of healthy donors were treated with PBS or 50 nM PMA for 4 h. **f**, Silver staining of samples from the EET protein isolation procedure. Lanes 1–2, supernatants of eosinophils treated with PBS (lane 1) or PMA (lane 2); lane 3, PBS washing buffer for PMA-treated eosinophils; lane 4, PMA-treated eosinophils incubated with medium without DNase I; lane 5, PBS-treated eosinophils incubated with medium containing DNase I; and lane 6, PMA-treated eosinophils incubated with medium containing DNase I (lane 6). The data represents $n = 3$ independent experiments. **g**, The proteins in lanes 4 and 6 of **f** were identified by LC-MS/MS. Proteins that were present in lane 6 and absent in lane 4 were compared for three donors. Those that overlapped in at least two donors were defined as EET-binding proteins. The representative EET-binding proteins are shown in the schematic. **h**, Relative levels of 16S and 18S DNA in EETs. Data are the mean $\pm$ s.d.; $n = 5$ independent experiments. **b, c, e**, Data are the mean $\pm$ s.d.; $n = 6$ mice per group, representative of $n = 2$ independent experiments; compared by one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test (**b, c**) or a two-tailed unpaired Student's *t*-test (**e**).

wild-type controls (Fig. 4e and Extended Data Fig. 4e), suggesting that *PAD4* knockout abolishes EET formation. More importantly, we found inhibition of goblet-cell hyperplasia and mucus production in the lungs of the *PAD4*^{-/-} mice (Fig. 4f,g). In addition, the *PAD4*^{-/-} mice displayed a reduced OVA-elicited immune response, as evidenced by decreased numbers of inflammatory cells and expression of type 2 cytokines (Fig. 4h,i). In addition, the levels of upstream cytokines—including IL-25, IL-33 and TSLP—in the

BALF also decreased in the *PAD4*^{-/-} mice compared with the wild types (Extended Data Fig. 4f). Given both NETs and EETs require *PAD4*, we adoptively transferred wild-type neutrophils, wild-type eosinophils or *PAD4*^{-/-} eosinophils to *PAD4*^{-/-} mice before OVA challenge. We found that the transfer of wild-type eosinophils, but not neutrophils or *PAD4*^{-/-} eosinophils, rescued the goblet-cell hyperplasia, mucus production and pulmonary inflammation that had been diminished in *PAD4*^{-/-} mice (Fig. 4e–i and Extended

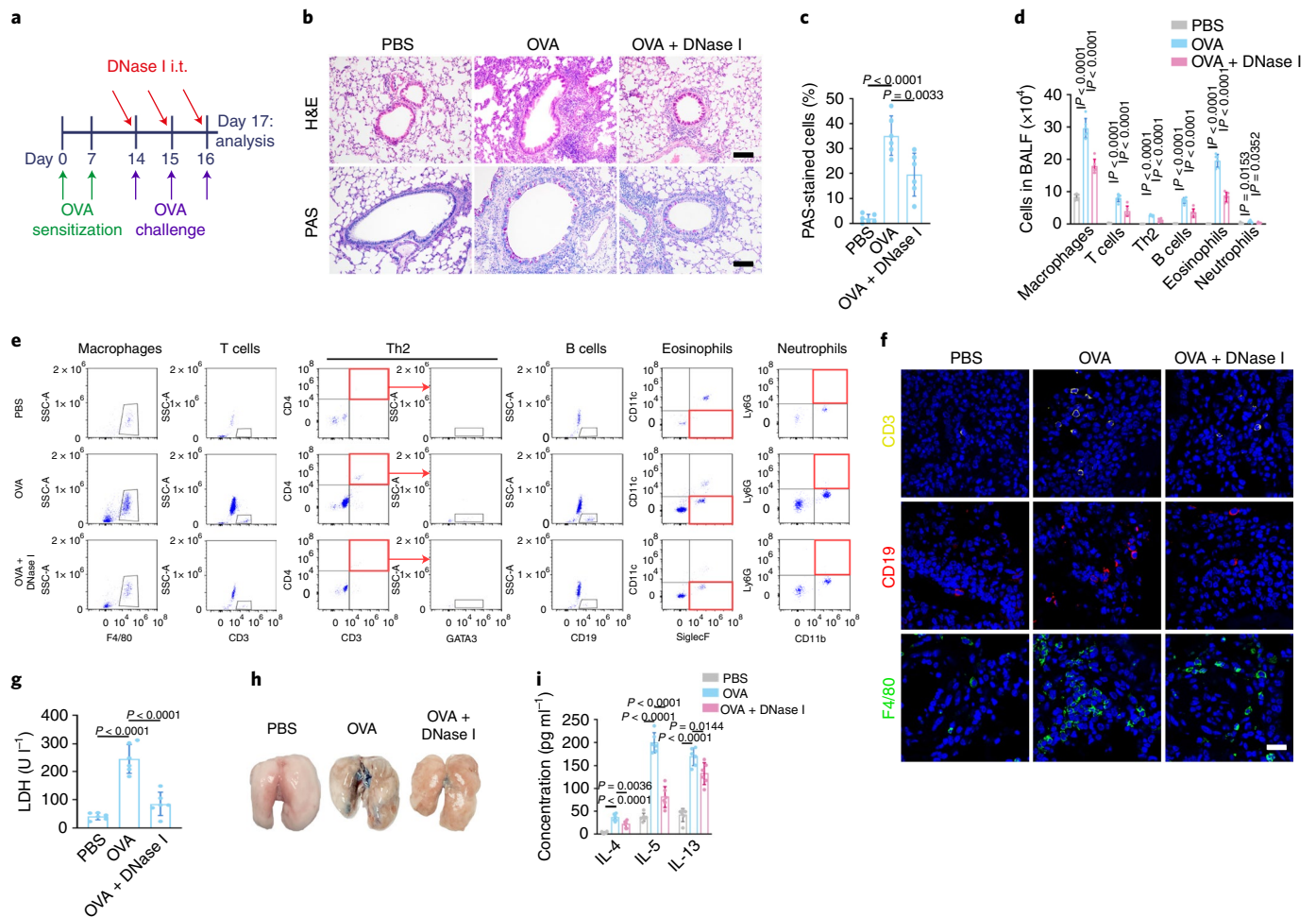


Fig. 3 | DNase I treatment ameliorates asthma in mice. C57BL/6 mice were sensitized and challenged with PBS or OVA. The OVA-sensitized mice were administered with DNase I intratracheally (i.t.) before being challenged with OVA. **a**, Experiment schematic. **b**, Representative images of haematoxylin and eosin (H&E; top) and periodic acid-Schiff (PAS; bottom) staining of lung sections of the mice. Scale bars, 100 μ m. **c**, The percentage of PAS-stained goblet cells in the total epithelial cells in the lung sections. **d**, **e**, Immune cells in the BALF were evaluated by flow cytometric analysis. The number of cells (**d**) and representative plots (**e**) are shown. **f**, Representative immunofluorescence images of CD3, CD19, F4/80 and DAPI staining of lung sections of the mice. Scale bar, 20 μ m. **g**, Levels of lactate dehydrogenase (LDH) in the BALF. **h**, The mice were injected with Evans blue dye via the tail-vein 1h before their lungs were harvested. Representative images of the lung tissues showing the accumulation of Evans blue. **i**, The levels of the indicated type 2 cytokines in the BALF were evaluated by ELISA. **c**, **d**, **g**, **i**, Data are the mean $\pm$ s.d.; $n=6$ mice per group, representative of $n=2$ independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test.

Data Fig. 4e,f), indicating that the effects of citrullinated extracellular trap deficiency in OVA-challenged mice are mainly attributed to EETs. Collectively, our data suggest that EETs boost the type 2 immune response and amplify asthma progression.

EETs aggravate allergic inflammation by activating PNECs. Next, we explored the potential mechanisms underlying the impact of EETs on asthma progression. We recently identified CCDC25 as a NET-DNA sensor on tumour cells¹⁵. Thus, we investigated whether CCDC25 plays a role in an EET-mediated asthma response. By analysing the expression profiles of CCDC25 in different tissues from online transcriptomic data^{15,37}, we found that tissues of the nervous system exhibited high levels of CCDC25 (Extended Data Fig. 5a). Moreover, the crosstalk between the immune and neuroendocrine systems plays an important role in allergic inflammation at mucosal surfaces^{38,39}. It raises the possibility that EETs may drive the lung allergic response through interactions with the neuroendocrine system.

Interestingly, we found that EETs in the lungs of OVA-immunized mice resided in close proximity to PNECs denoted as CGRP-positive^{19,40} (Fig. 5a). Quantitative analysis⁴¹ showed that the amount of EETs in proximity to PNECs was markedly higher than those localizing in the whole airway wall and those in proximity to blood vessels (Extended Data Fig. 5b). Moreover, immunofluorescence analysis showed that PNECs exhibited strong signals of CCDC25 (Fig. 5b and Extended Data Fig. 5c). Given that it is not feasible to isolate primary mouse PNECs, we used lung homogenates to detect CCDC25 (by western blotting) and observed that the levels of CCDC25 in the lung tissues of mice increased slightly after OVA challenge (Extended Data Fig. 5d).

To further explore the interaction between EETs and PNECs, we quantified the levels of neurotransmitters secreted by PNECs—that is, vasoactive intestinal peptide (VIP), GABA-synthesizing enzyme (GAD1, GABA-related gene), neuropeptide Y (NPY), chromogranin A (CHGA) and CGRP (CGRP)—in the lung tissues of the mice treated with OVA and DNase I. We observed that challenge

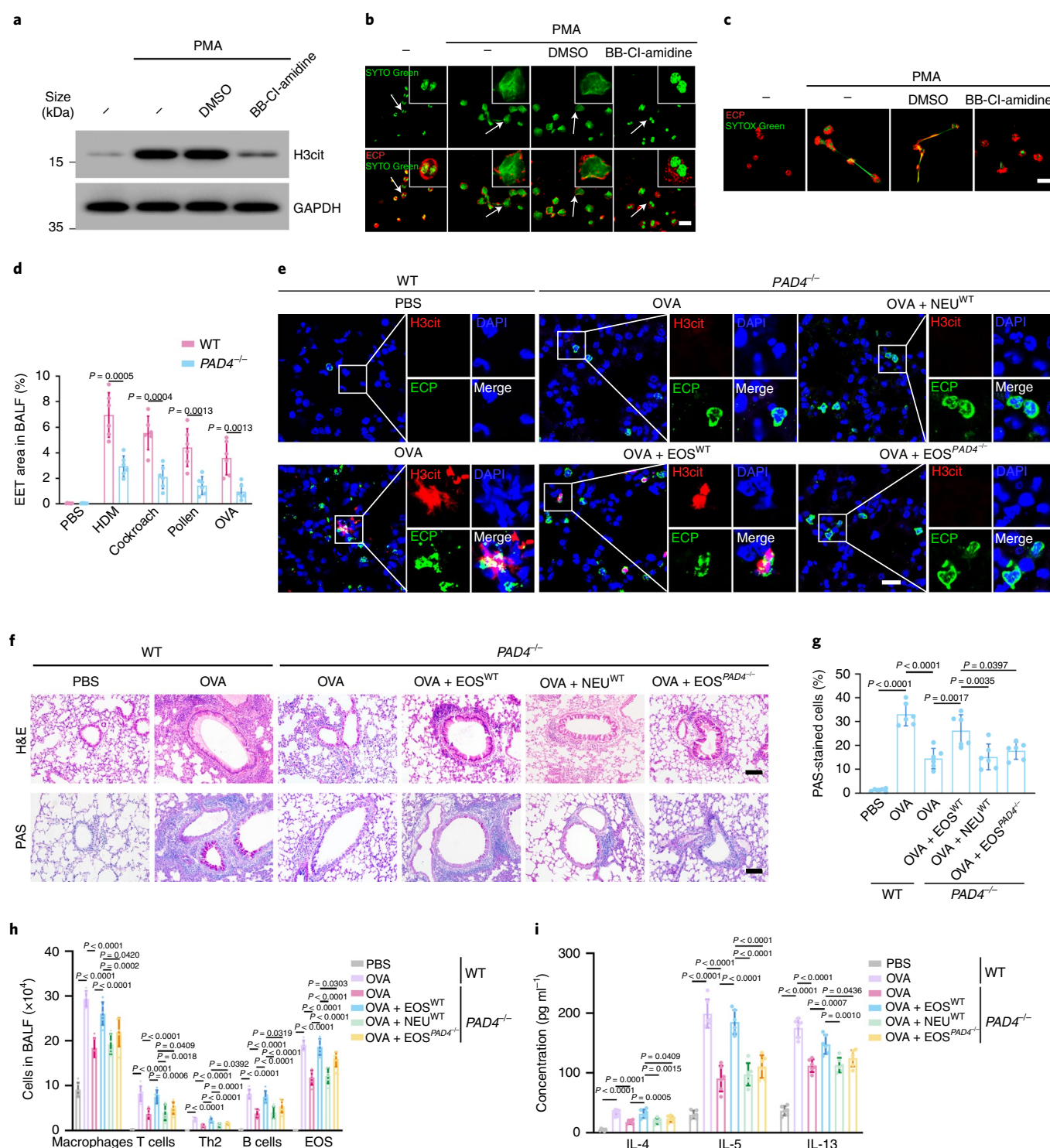


Fig. 4 | EETs amplify allergic asthma responses. **a–c**, Human eosinophils were pretreated with or without dimethylsulfoxide (DMSO) or BB-Cl-amidine and stimulated with 50 nM PMA for 4 h (ref. ⁷⁶; **a,c**) or 20 nM PMA for 2 h (ref. ⁷⁷; **b**). **a**, Representative immunoblots of H3cit and GAPDH in eosinophils from $n=3$ independent experiments. **b**, Chromatin morphology was examined by SYTO Green and ECP staining. The arrows point to the areas that are magnified in the insets. **c**, EET formation was examined by SYTOX Green and ECP staining. **d**, Wild-type C57BL/6 and $PAD4^{-/-}$ mice were immunized with PBS, HDM, cockroach, pollen or OVA. The level of EETs in the BALF smears was determined as a percentage of the EET area. **e–i**, Wild-type and $PAD4^{-/-}$ mice were sensitized and challenged with PBS or OVA. The $PAD4^{-/-}$ mice were adoptively transferred with 2×10^5 eosinophils (EOS) or neutrophils (NEU) isolated from wild-type or $PAD4^{-/-}$ mice. **e**, Representative immunofluorescence images of H3cit, ECP and DAPI staining of lung sections of the mice from the different treatment groups. **b,c,e**, Scale bars, 20 μ m. **f**, Representative images of H&E (top) and PAS (bottom) staining of the lung sections. Scale bar, 100 μ m. **g**, Quantification of **f**. **h**, Immune cells in the BALF were quantified by flow cytometric analysis. **i**, The levels of the indicated type 2 cytokines in the BALF were evaluated by ELISA. **d,g–i**, Data are the mean $\pm$ s.d.; $n=6$ mice per group, representative of $n=2$ independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test (**g–i**) or a two-tailed unpaired Student's t -test (**d**). WT, wild type.

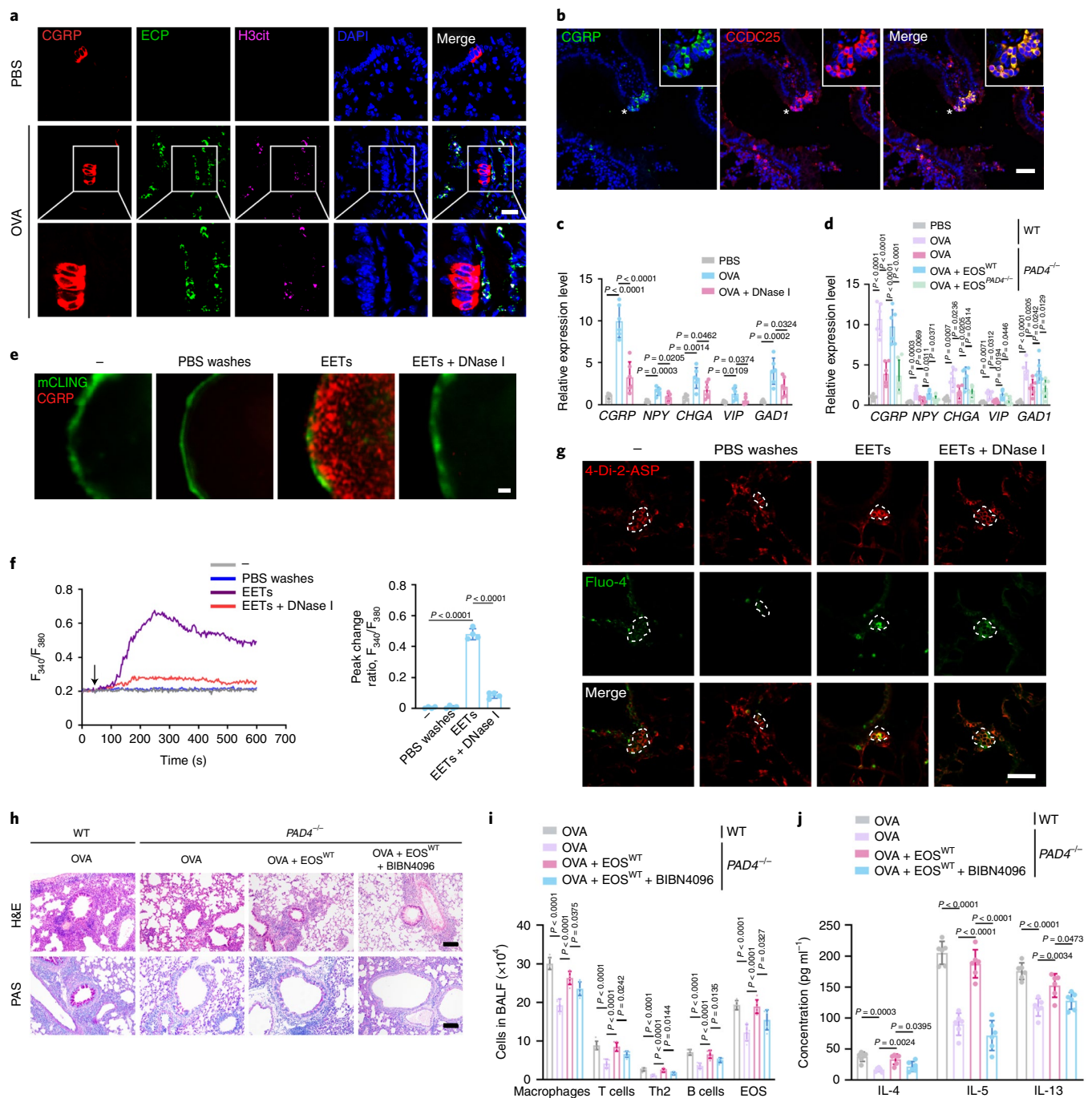


Fig. 5 | EETs aggravate allergic inflammation by activating PNECs. **a**, Representative immunofluorescence images of CGRP, ECP, H3cit and DAPI staining of lung sections of mice immunized with PBS (top) or OVA (bottom). Scale bar, 20 μm . **b**, Representative immunofluorescence images of CGRP, CCDC25 and DAPI staining of mouse lung sections ($n=6$). The asterisks indicate the area shown at a higher magnification in the insets. Scale bar, 50 μm . **c,d**, Wild-type and $PAD4^{-/-}$ mice were immunized with OVA. **c**, The wild-type mice were treated with or without DNase I before being challenged with OVA. **d**, The $PAD4^{-/-}$ mice were transferred with the indicated eosinophils. **c,d**, The expression levels of genes encoding neuropeptides (CGRP, NPY, CHGA and VIP) and GABA-synthesizing enzyme (GAD1) in lung homogenates were examined by RT-qPCR. **e,f**, H146 cells were treated with or without the PBS washes and EETs in the presence or absence of DNase I. **e**, The H146 cells were transfected with EYFP-CGRP plasmid before the indicated treatments. Representative fluorescence images of EYFP-CGRP expression in H146 cells stained with mCLING. Scale bar, 1 μm . **f**, Cytosolic Ca^{2+} mobilization in H146 cells was examined using Fura-2/AM ratiometric fluorescence measurements. Fluorescence curves (left) and peak responses (right) are shown. The arrow indicates the time of stimulation. **g**, Vibratome slices of living lungs loaded with Fluo-4 were treated without or with the PBS washes or EETs in the presence or absence of DNase I. PNECs, indicated by the dotted lines, were visualized by 4-Di-2-ASP staining. Representative images of slices 240 s after the stimulation are shown ($n=3$). Scale bar, 50 μm . **h-j**, Wild-type and $PAD4^{-/-}$ mice were immunized with OVA. Wild-type eosinophils were adoptively transferred to the $PAD4^{-/-}$ mice with or without BIBN4096 treatment. **h**, Representative images of H&E (top) and PAS (bottom) staining of lung sections. Scale bar, 100 μm . **i**, Immune cells in the BALF were quantified by flow cytometric analysis. **j**, The levels of the indicated type 2 cytokines in the BALF were evaluated by ELISA. **c,d,f,i,j**, Data are the mean $\pm$ s.d.; $n=6$ mice per group, representative of $n=2$ (**c,d,i,j**) and 4 (**f**) independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test. WT, wild type.

with OVA increased the expression of neurotransmitters, which was markedly reduced by intratracheal DNase I administration (Fig. 5c). In agreement with this, *PAD4* knockout suppressed the expression of neurotransmitters induced by OVA, which could be rescued by the transfusion of wild-type but not *PAD4*^{-/-} eosinophils (Fig. 5d). Among the neurotransmitters, CGRP and GABA-synthesizing enzyme were more prominently upregulated by OVA treatment (Fig. 5c,d).

To explore the role of CGRP or GABA in asthmatic inflammation, we administered the CGRP receptor antagonist (BIBN4096) and GABA receptor antagonist (CGP35348) to OVA-treated mice. We found that BIBN4096 was much more effective in suppressing the inflammatory-cell infiltration (Extended Data Fig. 5e), type 2-cytokine release (Extended Data Fig. 5f) and mucus secretion (Extended Data Fig. 5g) elicited by OVA, suggesting that CGRP is a major neuropeptide that exacerbates asthma.

To explore whether EETs directly induce neuropeptide release, we isolated EETs from the culture of eosinophils stimulated with PMA as we previously described¹⁵. We applied EETs to H146 cells, which have neuroendocrine functions^{42,43}. Quantitative PCR revealed that EETs, and not the PBS-wash control prepared in the absence of eosinophils, induced CGRP overexpression in H146 cells (Extended Data Fig. 5h). We transfected H146 cells with enhanced yellow fluorescent protein (EYFP)-tagged CGRP plasmid to visualize the neuropeptide production. Using stimulated emission-depletion super-resolution microscopy, we observed overexpression of EYFP-labelled CGRP in the H146 cells after the addition of EETs (Fig. 5e and Extended Data Fig. 5i). Moreover, EETs increased the concentration of cytosolic Ca²⁺ in the H146 cells (Fig. 5f). To investigate whether EETs activate primary PNECs, we employed the lung vibratome section model for live-cell imaging. Stimulation of EETs led to a rise in the cytosolic Ca²⁺ concentration of PNECs (Fig. 5g and Extended Data Fig. 5j). Importantly, DNase I treatment dramatically reversed these effects (Fig. 5e–g and Extended Data Fig. 5h–j).

To further confirm that EETs aggravate allergic inflammation by stimulating the release of CGRP, we transferred wild-type eosinophils to OVA-sensitized *PAD4*^{-/-} mice and concomitantly administered BIBN4096. We found that the goblet-cell hyperplasia, mucus production and pulmonary inflammation in *PAD4*^{-/-} mice restored by wild-type eosinophils was significantly inhibited by the BIBN4096 treatment (Fig. 5h–j and Extended Data Fig. 5k). Collectively, these findings suggest that EETs induce the secretion of neuropeptides and neurotransmitters by PNECs, which exacerbates asthmatic inflammation.

EETs activate PNECs via CCDC25. To determine whether CCDC25 was responsible for the EET-mediated activation of PNECs, we transfected short interfering RNA (siRNA) against *CCDC25* into H146 cells with CGRP forced expression. We then added a dilution series of EETs isolated from PMA-treated eosinophils. We observed that EETs boosted CGRP expression in a dose-dependent manner, which was inhibited by CCDC25 knockdown (Fig. 6a and Extended Data Fig. 6a). Moreover, we knocked out *CCDC25* in H146 cells using clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR associated protein 9 (Cas9). In corroboration with the siRNA and reporter approaches, quantitative PCR with reverse transcription (RT–qPCR) revealed that the endogenous expression of CGRP induced by EETs was abrogated (Fig. 6b and Extended Data Fig. 6b). Moreover, we treated H146 cells with a rabbit anti-human polyclonal antibody against CCDC25 (ref.¹⁵) and found that the EET-induced CGRP overexpression was significantly reduced (Fig. 6c and Extended Data Fig. 6c).

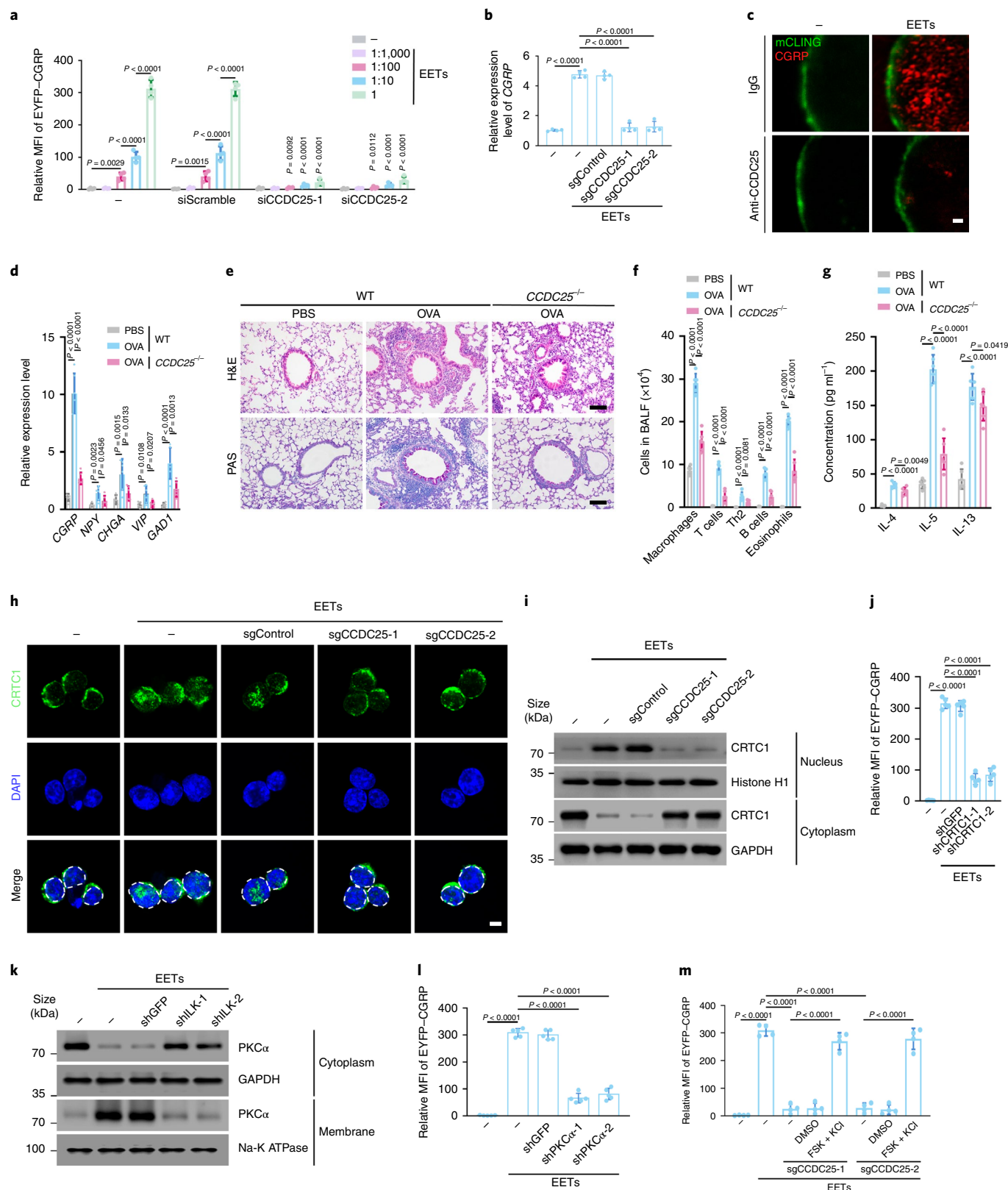
To corroborate these findings in vivo, we applied OVA to mice deficient in CCDC25 (*CCDC25*^{-/-})¹⁵. The release of neuropeptides and neurotransmitters elicited by OVA markedly decreased in the *CCDC25*^{-/-} mice compared with the wild-type controls (Fig. 6d). In parallel to the changes of neuroendocrine factors, the *CCDC25*^{-/-} mice had diminished goblet-cell hyperplasia and mucus production in their lungs in comparison to the wild-type mice (Fig. 6e and Extended Data Fig. 6d). Infiltrating inflammatory cells and type 2 cytokines were also significantly reduced in the *CCDC25*^{-/-} mice (Fig. 6f,g and Extended Data Fig. 6e). Notably, neither eosinophil recruitment nor EET production was reduced in the *CCDC25*^{-/-} mice at the early time point of the OVA challenge (day 15). In contrast, eosinophil recruitment and EET production were significantly diminished in the *CCDC25*^{-/-} mice at the late time point (day 17; Extended Data Fig. 6f,g).

It has been reported that the nuclear translocation of CREB-regulated transcriptional coactivator 1 (CRTC1), which is initiated by a calcium-signalling cascade, promotes CGRP expression in neurons⁴⁴. Given that EETs activate calcium influx in PNECs, we questioned whether CRTC1 has any effect on CGRP overexpression induced by EETs. We observed that stimulation of EETs resulted in CRTC1 shuttle to the nucleus of H146 cells, which was abrogated by *CCDC25* knockout (Fig. 6h,i). More importantly, transduction of short hairpin RNA (shRNA) to *CRTC1* significantly reduced CGRP expression (Fig. 6j and Extended Data Fig. 6h). These data suggest that EET–CCDC25 signals initiate calcium influx and consequently enhance CGRP expression via CRTC1 nuclear translocation. It has been reported that activation of ILK, which interacts with CCDC25 (ref.¹⁵), results in the plasma

Fig. 6 | EETs activate PNECs via CCDC25. **a,c,j,l,m**, H146 cells were transfected with an EYFP-tagged CGRP plasmid. **a**, H146 cells were transfected with siRNA targeting CCDC25 (siCCDC25) and treated with EETs at different dilutions. The relative mean fluorescence intensity (MFI) of the EYFP–CGRP signal was quantified by fluorescence imaging. Data are the mean ± s.d.; *n* = 4 independent experiments; data in the siCCDC25 groups were compared with the same dilution of the first group; compared by two-way ANOVA with Sidak's multiple comparisons test; siScramble was used as a negative control. **b**, H146 cells with *CCDC25* knockout (using single guide RNA (sgRNA) to *CCDC25*, sgCCDC25) were treated with EETs. The expression levels of CGRP were quantified by RT–qPCR; sgControl, control sgRNA. **c**, Representative fluorescence images of EYFP–CGRP in H146 cells expressing EYFP–CGRP after the indicated treatments. Scale bar, 1 μm. **d–g**, Wild-type and *CCDC25*^{-/-} mice were immunized with OVA. **d**, Expression levels of the indicated genes, determined by RT–qPCR, in lung homogenates. **e**, Representative images of H&E and PAS staining of lung sections of the mice. Scale bars, 100 μm. **f**, Immune cells in the BALF were quantified by flow cytometry. **g**, The levels of the indicated cytokines in the BALF were evaluated by ELISA. **h,i**, H146 cells with *CCDC25* knockout were stimulated with EETs. **h**, Representative images of CRTC1 and DAPI staining of H146 cells (*n* = 3). The dotted lines indicate the nucleus area. Scale bar, 5 μm. **i**, Representative CRTC1 immunoblots of the nuclear and cytoplasm fractions (*n* = 3). **j**, The relative MFI of the EYFP–CGRP signals in the H146 cells following the indicated treatments was quantified by fluorescence imaging; shGFP, shRNA to GFP; shCRTC1, shRNA to CRTC1. **k**, Representative immunoblots of PKCα in the membrane and cytoplasm of H146 cells following the indicated treatment (*n* = 3); shILK, shRNA to ILK. **l**, The relative MFI of the EYFP–CGRP signals in the H146 cells following the indicated treatments was quantified by fluorescence imaging; shPKCα, shRNA to PKCα. **m**, *CCDC25* was knocked out of H146 cells. After EET stimulation in the presence or absence of forskolin (FSK) and KCl, the relative MFI of the EYFP–CGRP signals in the H146 cells were quantified by fluorescence imaging. **b,d,f,g,j,l,m**, Data are the mean ± s.d.; *n* = 6 mice per group, representative of *n* = 2 (**d,f,g**), 4 (**b,m**) and 5 (**j,l**) independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test. WT, wild type.

membrane recruitment of PKC α ^{45,46}, which increases calcium influx^{47–49}. In agreement with this, we found that the administration of EETs to a H146 cell culture induced membrane translocation of PKC α , which was greatly reduced by the inhibition of both CCDC25 and ILK (Fig. 6k and Extended Data Fig. 6i,j). Furthermore,

silencing of PKC α reduced the calcium influx (Extended Data Fig. 6k,l), CRTCl nuclear translocation (Extended Data Fig. 6m) and CGRP expression (Fig. 6l) induced by EETs. Moreover, forced CRTCl nuclear translocation through treatment with forskolin and KCl⁴⁴ rescued the suppression of CGRP expression by CCDC25



knockout (Fig. 6m). Together, these data suggest that EETs activate PNECs via a *CCDC25*–ILK–PKC α –CRTC1 pathway.

EPX potentiates the interaction between EETs and PNECs. Granule proteins play an important role in the functions of NETs^{50,51}. To investigate whether eosinophil granule proteins have any effect on the EET–PNEC interaction, we isolated eosinophil granule proteins from primary human eosinophils as previously described^{52,53}. We observed that addition of eosinophil granule proteins did not have any appreciable effect on the expression of CGRP in H146 cells. In contrast, synthetic 8-hydroxy-2'-deoxyguanosine (8-OHdG)-modified DNA, which activates *CCDC25* (ref.¹⁵), and DNA extracted from eosinophils treated with PMA (EET-DNA) increased the CGRP levels (Fig. 7a). Although eosinophil granule proteins alone did not increase the levels of CGRP, the pretreatment of eosinophil granule proteins markedly enhanced the CGRP overexpression stimulated by both synthetic 8-OHdG-modified DNA and EET-DNA (Fig. 7a). This effect was *CCDC25*-dependent as it could be abrogated by *CCDC25* knockout (Fig. 7b).

Given our proteomic analysis showing that EETs contain multiple eosinophil granule proteins (Fig. 2g and Supplementary Table 1), we added their corresponding recombinant human proteins along with synthetic 8-OHdG-modified DNA to H146 cells. We found that eosinophil peroxidase (EPX), but not the other proteins, significantly increased CGRP expression induced by synthetic 8-OHdG-modified DNA (Fig. 7c). To investigate the effect of EPX on EET-mediated inflammation *in vivo*, we challenged mice deficient in EPX (*EPX*^{−/−}) with OVA. We found that mucus production (Fig. 7d and Extended Data Fig. 7a), inflammatory-cell infiltration (Fig. 7e), type 2-cytokine levels (Extended Data Fig. 7b) and CGRP expression (Extended Data Fig. 7c) were significantly decreased in the *EPX*^{−/−} mice compared with the wild types. However, *EPX* knockout did not have an appreciable effect on EET formation in the BALF (Fig. 7f and Extended Data Fig. 7d). These effects were specific as the CGRP expression levels and airway inflammation were comparable between wild-type mice and mice deficient in eosinophil major basic protein (MBP; Fig. 7d,e and Extended Data Fig. 7a–c), another eosinophil granule protein identified in EETs. Moreover, we transferred eosinophils from wild-type and *EPX*^{−/−} mice into *PAD4*^{−/−} mice. Our data showed that CGRP production (Fig. 7g) and airway inflammation (Extended Data Fig. 7e,f) in *PAD4*^{−/−} mice injected with *EPX*^{−/−} eosinophils was lower than the mice that were injected with wild-type eosinophils.

Moreover, we delivered dsDNA purified from splenocytes or EETs of wild-type, *PAD4*^{−/−} and *EPX*^{−/−} eosinophils into wild-type and *CCDC25*^{−/−} mice. Both DNA from EETs and splenocytes increased CGRP production (Fig. 7h). However, the effects of the

EET-DNA were stronger than those of the splenocyte DNA (Fig. 7h). This is probably because DNA from granulocyte extracellular traps contains higher levels of 8-OHdG modification¹⁰ and *CCDC25* preferentially binds to 8-OHdG-enriched DNA¹⁵. Moreover, *PAD4* knockout rather than *EPX* knockout reduced CGRP expression induced by EET-DNA (Fig. 7h), which is consistent with our data that the program for EET formation (EETosis) was not significantly diminished in *EPX*^{−/−} mice (Fig. 7f and Extended Data Fig. 7d). The effects of intratracheal DNA delivery were almost completely abolished in the *CCDC25*^{−/−} mice (Fig. 7h). Furthermore, we mixed EET-DNA with recombinant EPX protein and found that EPX significantly increased CGRP expression induced by EET-DNA (Fig. 7h), which corroborated the *in vitro* results (Fig. 7c).

Discussion

Here we have identified an important role of EETs in integrating immunological and neurological signals to promote the development of severe asthma. Our data show that PNECs detect EET-DNA via *CCDC25* and subsequently fuel allergic inflammation by secreting neuroendocrine factors (Fig. 7i). In summary, our study uncovers a crucial role of EETs in neuro-immune crosstalk and provides a feasible target to ameliorate asthma.

The emerging field of extracellular nucleic acid signals is one of the most exciting frontiers in biomedical research; NETs released from neutrophils is the central spotlight of this fast-moving area. The immune-modulatory functions of NETs have been implicated in the pathology of a growing number of diseases, including inflammation⁵⁴, tumour¹⁵ and autoimmune diseases¹⁰. Previous studies have reported that NETs contribute to the pathogenesis of asthma induced by lipopolysaccharides¹¹ and rhinovirus²⁶, suggesting that NETs play an important role in asthma exacerbation triggered by acute bacterial or viral infection. Interestingly, it has been shown that significant dsDNA release can still be observed in uninfected HDM-exposed mice deficient in NETs²⁶, indicating that there is an alternative source of extracellular DNA in non-infectious asthma. Here we provide a possible explanation for these observations by demonstrating that EETs produced from eosinophils have a critical impact on the progression of non-infectious asthma.

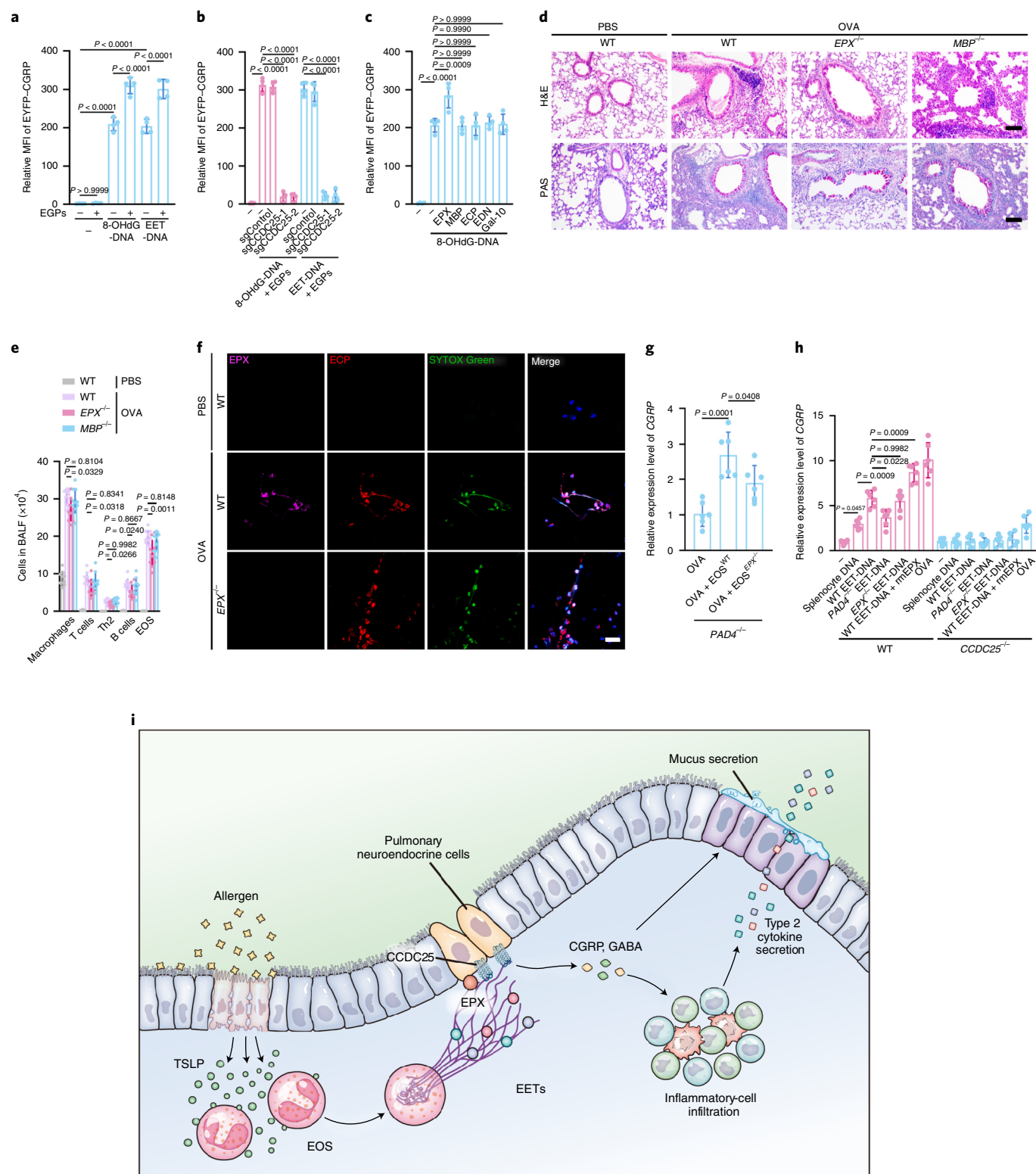
A key feature of asthma is airway inflammation, which is primarily orchestrated by type 2 immune responses⁵⁵. Eosinophils are one of the most abundant infiltrating inflammatory cells associated with type 2 immune responses^{3,4,29}. Clinically, eosinophilic airway inflammation is present in more than 50% of adults with asthma¹, whereas the prevalence of neutrophilic asthma is only 15%^{56,57}. In an analysis of endobronchial biopsy specimens from asthma patients, NETs were only present in about 20% of the patients, whereas peribronchial EETs could be found in all of the patients⁵⁸. Similarly, the

Fig. 7 | EPX potentiates the interaction between EETs and PNECs. **a**, H146 cells expressing EYFP–CGRP were incubated with or without 8-OHdG-DNA or EET-DNA in the presence or absence of eosinophil granule proteins. The relative MFI of the EYFP–CGRP signals in the H146 cells was quantified by fluorescence imaging. **b**, H146 cells expressing EYFP–CGRP were transduced with sg*CCDC25* and stimulated with 8-OHdG-DNA or EET-DNA in the presence of eosinophil granule proteins. The relative MFI of the EYFP–CGRP signals in the H146 cells was quantified by fluorescence imaging. **a,b**, EGPs, eosinophil granule proteins. **c**, H146 cells expressing EYFP–CGRP were treated with 8-OHdG-DNA in the presence of human recombinant EPX, MBP, ECP, eosinophil-derived neurotoxin (EDN) or galectin-10 (Gal-10). The relative MFI of the EYFP–CGRP signals in the H146 cells was quantified by fluorescence imaging. **d,e**, Wild-type, *EPX*^{−/−} and *MBP*^{−/−} mice were immunized with OVA. **d**, Representative images of H&E and PAS staining of lung sections of the mice. Scale bar, 100 μ m. **e**, Immune cells in the BALF were quantified by flow cytometry. **f**, Wild-type and *EPX*^{−/−} mice were immunized with OVA. Representative immunofluorescence images of EPX, ECP, SYTOX Green and DAPI staining of the BALF. Scale bar, 20 μ m. **g**, *PAD4*^{−/−} mice were adoptively transferred with eosinophils isolated from wild-type or *EPX*^{−/−} mice. **h**, Wild-type and *CCDC25*^{−/−} mice were intratracheally instilled with splenocyte dsDNA or EET-DNA from the indicated eosinophils with or without EPX. As positive controls, mice were immunized with OVA. rmEPX, recombinant mouse EPX. **g,h**, The CGRP expression levels of the lung homogenates of the mice were examined by RT–qPCR. **i**, Schematic summarising the major discoveries of this study. Allergens induce TSLP, which triggers the formation of EETs in asthmatic airways. The EET-DNA activates PNECs, which is potentiated by the eosinophil granule protein EPX. Subsequently, PNECs amplify type 2 immune responses via neuropeptides and neurotransmitters such as CGRP and GABA. **a–c,e,g,h**, Data are the mean $\pm$ s.d.; $n = 4$ independent experiments (**a–c**); $n = 6$ (**g,h**) or 10 (**e**) mice per group, representative of $n = 2$ independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test (**b,c,e,g,h**) or two-way ANOVA with Sidak's multiple comparisons test (**a**). WT, wild type; EOS, eosinophils.

inhalation of allergens by mice increases eosinophils in the lung lavage fluid but scarcely changes the quantity of neutrophils^{5,27,28}. In agreement with this, we observed that the number of eosinophils, but not neutrophils, markedly increase in mouse lung tissues and more than half of the SYTOX Green staining signals co-locate with ECP following OVA stimulation, indicating that EETs are much more abundant than NETs during non-infectious asthma.

Despite the much higher numbers of eosinophils in asthma, reports of EETs remain very limited compared with those of NETs.

Previous studies show that EETs are present in nasal polyps of patients with chronic rhinosinusitis³⁹ and EETs are more likely to be generated by eosinophils isolated from the peripheral blood of patients with more severe asthma^{60,61}. Here we demonstrated that EETs in the BALF are associated with the severity of asthma in patients. This is very clinically important because BALF samples are much more pathologically relevant to airway inflammation and asthma severity than blood samples. Moreover, it has been observed that EETs play an important role in asthma progression in mice⁶⁰.



However, the underlying mechanism has remained largely unclear. Here we demonstrate that extracellular DNA in EETs activates PNECs via the CCDC25–ILK–PKC α –CRTC1 pathway. PNECs are a critical source of neuropeptides and neurotransmitters in airways, which shape immune responses in asthma^{19,40,62,63}. In agreement with this, we found that PNECs amplify allergic inflammation via neuropeptides and neurotransmitters after activation by EETs. To our knowledge, the interaction between granulocyte extracellular traps and peripheral nerves is previously unreported.

Granulocyte extracellular traps consist of not only extracellular web-like DNA structures but also various intracellular proteins. It has been well documented that NET-binding proteins—including neutrophil elastase, MPO and matrix metalloproteinase 9—play an important role in various diseases^{9,12,64}. By contrast, much less is known about the functions or even components of EET proteins. Here we provide comprehensive EET-associated proteomics by LC–MS/MS and reveal that EETs are decorated with a number of granule proteins unique to eosinophils. Among them, EPX potentiates the interaction between EETs and PNECs. Interestingly, previous reports have shown that eosinophil granule proteins induce the hyperexcitability of airway nerves^{65–68}. Therefore, our study extends the current understanding of the neuro-immune interplay by showing that not only secretory eosinophil granule proteins but also those binding to extracellular DNA webs play an important role in mediating nerve signals.

We recently identified CCDC25 as a transmembrane DNA sensor that recognises NET-DNA and mediates liver metastasis¹⁵. However, its role in other non-tumoral diseases remains unknown. By analysing transcriptional data of CCDC25 in different tissues, we found that CCDC25 is highly expressed in the nervous system. An emerging role of neuro-immune interaction in asthma has been highlighted, partially due to the proximity between nerve fibres and immune cells in the respiratory mucosa^{69,70}. Eosinophils have been found closely apposed to nerve fibres and parasympathetic ganglia in airway inflammation^{71–73}. Our data reveal that PNECs, the only innervated airway epithelial cells, are surrounded by EETs in asthma tissues. PNECs are an important source of neuropeptides and neurotransmitters in lungs, which plays a crucial role in mediating allergic inflammation^{19,62,63}. Our data show that EETs enhance the endocrine function of PNECs by binding to CCDC25. Collectively, these data indicate that CCDC25 lies at the nexus between EETs and the peripheral nervous system, and may be a potential therapeutic target for asthma.

Beside granulocyte extracellular traps, extracellular DNA may be derived from other dead bystanders, such as local lung cells. In agreement with a previous report²⁶, we found that dsDNA purified from splenocytes can also induce CGRP production and inflammation as EET-DNA. However, the program for EET formation (EETosis) differs from necrotic cells as follows. First, CCDC25 preferentially binds to oxidized DNA, which is enriched in granulocyte extracellular traps^{15,74}. Second, the eosinophil granule protein EPX potentiates the interaction between EETs and PNECs. Third, neuropeptides can reciprocally recruit more eosinophils⁷⁵. Together, these data indicate that EET-DNA plays a predominant role in this neuro-immune axis during the exacerbation of non-infectious asthma. Furthermore, NETs can enhance DC recruitment and aggravate infection-related asthma^{11,26}. Therefore, we and others collectively demonstrated that granulocyte extracellular traps play an essential role in asthma progression. Our data suggest that EETs amplify allergic asthma responses mainly by activating PNECs. However, as CCDC25 is not exclusively expressed in PNECs, we do not entirely exclude the contribution of CCDC25 in other cell types to asthmatic inflammation, which is an important subject for future studies.

In summary, we demonstrate that extracellular DNA from eosinophils drives the allergen-induced airway immune response by

interacting with the peripheral nervous system. Our study has not only refined the understanding of EETs in asthma, but also established a previously unappreciated neuroimmunological circuit in responding to environmental stimuli.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41556-021-00762-2>.

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Methods

Human BALF samples. Asthma was diagnosed based on the GINA guidelines². Human BALF samples were obtained by fibre-optic bronchoscopy from three healthy volunteers (one male and two females) with an average age of 27.7 years, and 46 asthma patients (21 males and 25 females) with an average age of 29.3 years from Sun Yat-Sen Memorial Hospital. The FEV₁% (percentage of the vital capacity that the patient could expire in the first second of forced expiration to the full, forced vital capacity)⁷⁸ was obtained through a pulmonary function test. The classification level of asthma was assessed according to the Asthma Management section of the GINA guidelines². All samples were collected with informed consent and all of the participants received participant compensation. All related procedures were performed with the approval of the Internal Review and Ethics Boards of Sun Yat-Sen Memorial Hospital.

Scanning electron microscopy. Human BALF samples were collected and centrifuged at 1,000g for 10 min at 4°C. The samples were processed for scanning electron microscopy^{15,79}. Briefly, BALF sediments were dispensed onto glass slides, fixed with 2.5% glutaraldehyde (G5882, Sigma-Aldrich) and incubated with 1% osmium tetroxide. After dehydration with graded ethanol, the samples were desiccated, covered with 2 nm platinum and coated with 5 nm carbon. Images were acquired using an FEI QUANTA 200 scanning electron microscope.

Immunofluorescence. Human BALF sediments were smeared on glass slides and air dried. The samples were fixed with 4% paraformaldehyde and washed with PBS. After blockade, the samples were sequentially incubated with anti-ECP (MA5-34931, ThermoFisher Scientific), Alexa Fluor secondary antibody (cat. no. A31570, ThermoFisher Scientific) and DAPI (cat. no. D3571, ThermoFisher Scientific).

Mouse lungs were perfused with saline through the right ventricle and collected⁸⁰. The lungs were fixed with 4% paraformaldehyde and paraffin-embedded. Immunofluorescent staining of sections was then performed^{26,81}. Briefly, the lung sections were deparaffinized, rehydrated and antigen retrieved. After blockade, the sections were incubated overnight with antibodies to ECP (MBS2090412, MyBioSource), H3cit (ab5103, Abcam), CD3 (MAB4841, R&D), CD19 (3574S, Cell Signaling Technology), F4/80 (30325, Cell Signaling Technology), CGRP (PA1-85250, ThermoFisher Scientific) or CCDC25 (21209-1-AP, Proteintech), followed by incubation with secondary antibodies (A21202, A31571, A31573, A21206, A31572, A21208, A11055 and A21432; ThermoFisher Scientific). For the detection of dead lung epithelial cells, the samples were incubated with anti-mouse pan cytokeratin (ab9377, Abcam) at 4°C overnight, followed by incubation with secondary antibody (A31572, ThermoFisher Scientific) and DAPI (cat. no. D3571, ThermoFisher Scientific). After washing with PBS, the slices were incubated with 30 µl TUNEL reaction mixture (11684795910, Roche) for 30 min at 37°C. In the images, the proportion of TUNEL⁺CK⁺ cells in the total CK⁺ cells was divided by that of the PBS-treated mouse to quantify the dead epithelial cells.

For immunofluorescent staining of mouse BALF, BALF sediments were stained with 0.03 µM SYTOX Green (S7020, ThermoFisher Scientific), washed and smeared on glass slides. After fixation and blockade, the cells were incubated with antibodies to ECP (MBS2090412, MyBioSource), MPO (AF3667, R&D Systems) and EPX (orb382913, Biorbyt), followed by incubation with secondary antibody (A31570, A21432 and A31573; ThermoFisher Scientific) and DAPI (cat. no. D3571, ThermoFisher Scientific).

The level of EETs in human BALF was determined as the percentage of extracellular deposition of DNA and ECP²² using Imaris 9.0 (Bitplane)⁸¹. The nucleus was recognised using the DAPI channel. Spots were auto-generated to represent co-localized regions of extracellular DAPI and ECP. The area of EETs was calculated by manually subtracting the area of DAPI and ECP co-stained spots located inside the nuclei from the area of the entire spot.

The level of EETs in the BALF and lung sections of mice was determined as the percentage of the SYTOX Green⁺ECP⁺ and ECP⁺ H3cit⁺DAPI⁺ area, respectively, using Imaris 9.0. Briefly, 'co-localization result' channels were generated to represent the SYTOX Green⁺ECP⁺ or ECP⁺ H3cit⁺DAPI⁺ co-localized area using automatic thresholding module. The percentage of SYTOX Green⁺ECP⁺ or ECP⁺ H3cit⁺DAPI⁺ co-localized area was calculated using the 'Coloc function'.

For quantification of the proximity of EETs and PNECs, EETs and PNECs were identified as H3cit⁺ECP⁺ DAPI⁺ areas and CGRP⁺ cells, respectively, using Imaris 9.0 (refs. ^{81,82}). Briefly, a new channel representing the H3cit⁺ECP⁺ DAPI⁺ signal was built using the Coloc function. A circle with a 50 µm radius was drawn around the PNECs and set as the surface within 50 µm of the PNECs. The 0.1 µm-diameter spot of the H3cit⁺ECP⁺ DAPI⁺ channel was generated inside the surface to represent the EET area in proximity to PNECs. H&E staining for serial sections was performed to identify blood vessels and airway walls (from epithelium to serosa, excluding the cartilage and the 50 µm area surrounding the PNECs)^{41,72}. Randomly selected blood vessels and airway walls were manually delineated. The density of the H3cit⁺ECP⁺ DAPI⁺ spots (number of spots per mm²) within 50 µm of the PNECs, blood vessels and airway walls were calculated.

To quantify the expression levels of CCDC25 in PNECs and non-PNECs, the MFI of CCDC25 was compared between CGRP⁺ and CGRP⁻ cells using Imaris 9.0. Images with PNECs were acquired in randomly selected lung sections of control

mice. Spots representing CGRP⁺ or CGRP⁻ cells were generated in the CGRP and DAPI channel using the 'Spot function'. By manually delineating a 5 µm-radius surface for each spot, the area and CCDC25 intensity of the CGRP⁺ and CGRP⁻ cells were calculated using the 'Surface function'.

Smeared of H146 cells, neutrophils and eosinophils were incubated with antibodies to CRTCl (orb447387, Biorbyt), ECP (MA5-34931, ThermoFisher Scientific), MPO (AF3667, R&D Systems) or PAD4 (GTX113945, GeneTex) at 4°C overnight after fixation, permeabilization and blockade. This was followed by incubation with secondary antibodies (A21206, A31570 and A11058; ThermoFisher Scientific) and DAPI (cat. no. D3571, ThermoFisher Scientific)⁸³. Immunofluorescence images were acquired using an LSM 800, LSM 880 with Airyscan (Carl Zeiss) or a TCS SP8 STED super-resolution microscope (Leica).

Human peripheral blood cell isolation. Using density gradient centrifugation, granulocytes and mononuclear cells were isolated from the peripheral blood of healthy donors with Ficoll¹⁵. After treatment with RBC lysis buffer (420301, BioLegend), eosinophils were purified using a Human eosinophil isolation kit (130-092-010, Miltenyi Biotec). Monocytes, T cells, B cells and natural killer cells were purified using CD14 MicroBeads (130-050-201, Miltenyi Biotec) and a Pan T cell isolation kit (130-096-535, Miltenyi Biotec), B cell isolation kit II (130-091-151, Miltenyi Biotec) and NK cell isolation kit (130-092-657, Miltenyi Biotec), respectively. Neutrophils were purified using an EasySep direct human neutrophil isolation kit (19666, STEMCELL).

In vitro induction and purification of EETs. Human eosinophils (2 × 10⁵ per well) were seeded in 24-well culture plates that had been incubated with poly-L-lysine. After being treated with or without DMSO or 50 µM BB-Cl-amidine (HY-111347A, MCE) for 30 min, the eosinophils were treated with 50 nM PMA (P6741, Solarbio) for 4 h at 37°C to form EETs^{6,84}.

For the detection of EETs, samples were sequentially incubated with 0.03 µM SYTOX Green (S7020, ThermoFisher Scientific) and AF555-anti-ECP (bs-1754R-AF555, Bioss). The level of EETs was quantified by calculating the proportion of SYTOX Green⁺ECP⁺ areas in randomly selected fields using Imaris 9.0. To detect the level of H3cit, the samples were collected after PMA stimulation and subjected to immunoblotting.

For the purification of EETs, EETs adhered to the bottom of the culture plates were gently washed twice by incubation with PBS at 37°C for 10 min after the PMA-containing supernatant was removed. The EETs were then intensively washed down by pipetting with 50 µl of cold 0.9% NaCl (for the stimulation of vibratome sections) or DMEM medium with 10% fetal bovine serum (FBS; for the stimulation of H146 cells)¹⁵. After centrifugation, the supernatant containing EETs was collected¹⁵. In some experiments, we added PMA in the media without eosinophils and used the final PBS washes as the preparing control (PBS washes). Purified EET-containing solution induced from 2 × 10⁵ cells was used to treat H146 cells (1 × 10⁵ cells in each well) and mouse lung vibratome sections (50 µl in 0.9% NaCl). In some experiments, the EETs were diluted 1:10, 1:100 or 1:1,000.

For the extraction of EET-DNA, EETs induced from human or mouse eosinophils were isolated as described above. Cell-free supernatant containing EETs was incubated with proteinase (P4630, Sigma-Aldrich) for 4 h at 56°C (ref. ¹⁰). DNA was extracted from the supernatants using a MicroElute DNA clean up kit (D6296, Omega Bio-tek), resolved in sterile water and stored at -80°C. The EET-DNA isolated from human and mouse eosinophils was used for the qualitative analysis or stimulation of H146 cells and intratracheal administration of mice, respectively.

Animal experiments. *PAD4*^{fllox/fllox} (B6(Cg)-Padi4tm1.2Kmw/J, 026708) and *Ddx4-cre* (B6.FVB-Tg (Ddx4-cre)1Dcas/KnwJ, 018980) mice were purchased from the Jackson Laboratory. *PAD4*^{-/-} and *CCDC25*^{-/-} mice were generated by Shanghai Model Organisms Center¹⁵. Briefly, *PAD4*^{fllox/fllox} mice were crossed with *Ddx4-cre* mice to generate *PAD4*^{-/-} mice, which were then intercrossed to generate *PAD4*^{-/-} mice. *CCDC25*^{-/-} mice with a C57BL/6 genetic background were generated by deleting the genomic DNA fragment covering exon 3 of *CCDC25* using a CRISPR-Cas9-mediated genome-editing system. *EPX*^{-/-} (C57BL/6-*Epex*^{em3(cres)Smoc}/NM-KO-190794) and *MBP*^{-/-} (*Prg2*^{-/-}; C57BL/6-*Prg2*^{em1Smoc}/NM-KO-190857) mice were purchased from the Shanghai Model Organisms Center, Inc. Female wild-type C57BL/6 mice (2–8 weeks old) were purchased and maintained at the Animal Experiment Center of Sun Yat-Sen University under a cycle of 14 h lights on and 10 h lights off, and at 22°C and 60% humidity.

To generate a mouse model of asthma, mice were intraperitoneally sensitized with 10 µg OVA (A5503-5G, Sigma-Aldrich), HDM (XPB70D3A2.5, Greer), cockroach (RMB46P, Greer) or pollen (ab226242, Abcam) in 100 µl PBS emulsified with 100 µl of alum adjuvant (cat. no. 77161, ThermoFisher Scientific) on days 0 and 7 (ref. ⁸⁵). The mice were anaesthetized and intratracheally challenged with 100 µg OVA, HDM, cockroach or pollen in 50 µl PBS on days 14, 15 and 16. In some experiments, the mice were intratracheally treated with 500 IU DNase I (89836, ThermoFisher Scientific) 8 h before the allergen challenge²⁶. As a positive control for EET formation, the mice were intratracheally instilled with 20 µg PMA in 50 µl PBS 4 h before harvest. The mice were euthanized 24 h after the last allergen challenge.

For BALF collection, the mice were sacrificed with an overdose of pentobarbital sodium. After the trachea of the mice were exposed, a venous catheter was plugged in the cervical trachea and tied with a suture. The lungs were then consecutively flushed three times with 1 ml PBS and the infused fluid was collected with a syringe^{86,87}.

For the neutralization of IL-4, IL-5, IL-13, IL-25, IL-33 and TSLP, the mice were treated intraperitoneally with anti-IL-4 (14-7041-85, ThermoFisher Scientific; 0.125 mg per mouse), anti-IL-5 (14-7052-81, ThermoFisher Scientific; 0.125 mg per mouse), anti-IL-13 (16-7135-85, ThermoFisher Scientific; 0.075 mg per mouse), anti-IL-33 (AF-3626, R&D Systems; 0.075 mg per mouse), anti-IL-25 (MAB13992, R&D Systems; 0.5 mg per mouse), anti-TSLP (MAB555, R&D Systems; 0.025 mg per mouse) or IgG (MAB006R, R&D Systems; 0.075 mg per mouse) in 50 µl of PBS 1 h before each OVA challenge⁸⁸.

To block the neuropeptides *in vivo*, mice were intraperitoneally injected with BIBN4096 (SML2426, Sigma-Aldrich; 30 mg kg⁻¹) or CGP35348 (C5851, Sigma-Aldrich; 100 mg kg⁻¹) 2 h after the OVA challenge^{88,90}.

For the adoptive transfer experiments, mouse eosinophils were isolated from female wild-type, *PAD4*^{-/-} and *EPX*^{-/-} mice⁹¹. Briefly, mouse bone marrow cells were collected with cold medium (2% FBS in PBS). After red blood cell lysis and cell depletion using a Mouse lineage cell depletion kit (130-090-858, Miltenyi Biotec), eosinophil progenitors were cultured in RPMI 1640 medium supplemented with 20% FBS, 100 ng ml⁻¹ SCF (250-03, Peprotech) and Flt3L (250-31L, Peprotech)^{92,93}. After 4 d, the culture media were removed and fresh media containing 10 ng ml⁻¹ IL-5 (215-15, Peprotech) were added. The media were replaced every 2 d. On day 10, >95% cells in the culture were eosinophils, which was assessed using Hematocolor set (1116740001, Merck Millipore) staining⁹⁴. Mouse neutrophils were isolated from vein blood using a Mouse neutrophil isolation kit (130-097-658, Miltenyi Biotec). The mouse neutrophil purity was >95%, which was evaluated by flow cytometry using anti-CD11b (101212, BioLegend) and anti-Ly6G (127606, BioLegend)⁹⁵. Purified cells (2 × 10⁶ cells per recipient) in 50 µl PBS were instilled intratracheally 2 h before each OVA challenge^{96,97}.

For DNA administration, EET-DNA was purified as described above. Splenocyte dsDNA was purified using a High pure PCR template preparation kit (11796828001, Roche Life) according to the manufacturer's instructions²⁶. Mice were intratracheally administered with 10 µg dsDNA or EET-DNA in 50 µl PBS with or without 15 µg EPX protein (abx653285, Abxexa Ltd) four times at 24 h intervals. The mouse lungs were harvested and subjected to RT-qPCR analysis 24 h after the final treatment. All experiments and procedures associated with mice were authorized by the Institutional Review Boards and Animal Care and Use Committees of Sun Yat-Sen University.

Flow cytometry analysis for EETs. To measure the level of EETs in mouse BALF by flow cytometry, BALF cells were blocked and stained with 0.01 µM SYTOX Green (S7020, ThermoFisher Scientific)^{98,99}. Briefly, after washing with PBS, the cells were sequentially incubated with anti-H3cit (ab5103, Abcam) and anti-ECP (MBS2090412, MyBioSource), followed by incubation with Alexa Fluor secondary antibodies (A21428 and A21235; ThermoFisher Scientific). The cells were washed and centrifuged after each incubation. The samples were analysed on a CytoFLEX flow cytometer (Beckman Coulter).

ELISA of ECP-DNA. The level of EETs in the BALF was quantified using a capture ELISA by measuring ECP-DNA^{15,100}. Briefly, the 96-well plates (anti-mouse ECP ELISA kit, abx156952, Abxexa Ltd), which were pre-coated with anti-ECP, were blocked and sequentially incubated with BALF supernatant at 37 °C for 2 h and peroxidase-labelled anti-DNA monoclonal antibody (Cell death detection ELISA kit, 11774425001, Roche) at 37 °C for 1 h. Each incubation was followed by PBS washes. The wells were then incubated with peroxidase substrate (11774425001, Roche) at 26 °C for 10 min. Absorbance at 450 nm was detected using a microplate reader (infinite M200 PRO).

ELISA. Mouse BALF was sequentially centrifuged at 350g and 1,500g for 10 min. Cytokines in the supernatant were detected by anti-mouse IL-4 (M4000B, R&D Systems), IL-5 (M5000, R&D Systems), IL-13 (M1300CB, R&D Systems), IL-25 (DY1399, R&D Systems), IL-33 (M3300, R&D Systems) and TSLP (MTLP00, R&D Systems) ELISA kits according to the manufacturer's instructions.

EET protein purification. EET proteins were purified as previous studies on NETs, with slight modifications³³. Briefly, eosinophils from three donors were suspended in RPMI 1640 medium without phenol red and seeded in 24-well plates with poly-L-lysine-covered slides at a density of 1.7 × 10⁶ cells ml⁻¹ (0.85 × 10⁶ per well). The eosinophils were treated with PBS or 50 nM PMA for 4 h at 37 °C. The supernatant was removed and the wells were gently washed. To detach the EETs, the wells were incubated with DNase I (10 U ml⁻¹) for 20 min at 37 °C and stopped with 5 mM EDTA. As controls, EETs were incubated with RPMI 1640 without DNase I and PBS-treated eosinophils were administered with RPMI 1640 containing DNase I. The supernatant, washing buffer and media incubated with eosinophils were collected and sequentially centrifuged at 350g for 10 min and 16,000g for 20 min.

Silver staining. Silver staining was performed using samples prepared from eosinophils treated with or without PMA and digested with DNase I as described in the 'EET protein purification' section^{101,102}. Briefly, the samples were incubated with NuPAGE lithium dodecyl sulfate sample buffer (NP0007, ThermoFisher Scientific) and fractionated by 10% SDS-PAGE. The gel was stained using a ProteoSilver silver stain kit (PROTSIL1, Sigma-Aldrich) according to the manufacturer's instructions.

LC-MS/MS. EET proteins and controls were analysed by LC-MS/MS¹⁰³⁻¹⁰⁵. Briefly, the samples isolated from human eosinophils were sequentially treated with TCEP solution (646547, Sigma-Aldrich) and MMTS solution (208795, Sigma-Aldrich). After ultrafiltration, the samples were sequentially incubated with urea solution (pH 8.5) and TEAB solution (18597, Sigma-Aldrich), each followed by centrifugation at 12,000g for 20 min. The supernatants were incubated with TEAB solution and 2% trypsin (V5280, Promega) overnight and centrifuged to collect the supernatants, followed by vacuum drying to concentrate the peptides. The peptides were resolved with 0.1% formic acid and centrifuged. The supernatants were loaded onto Acclaim PepMap 100 C18 HPLC columns (160454 and 160321; ThermoFisher Scientific). The peptides were then analysed using a ThermoFisher Scientific Q Exactive mass spectrometer. For the MS scan, the resolution was set to 70,000 with a scan range of 350–1,800 *m/z*. For the top 20 ions, the MS2 spectra were acquired with a resolution of 15,000.

For the EET protein analysis, raw data were analysed using the MassMatrix File Conversion software version 3.9 and aligned with the UniProt *Homo sapiens* database (downloaded in April 2021; 20,396 entries). Proteins identified in the control groups were subtracted from the EET samples. The criterion for EET protein was defined as proteins only found in EET samples from at least two donors.

Qualitative analysis for EET-DNA. To evaluate the origin of EET-DNA, we quantified the relative abundance of mitochondrial and chromosomal DNA in EET-DNA using RT-qPCR¹⁰. Isolated DNA was mixed with SYBR Green master mix kit (4309155, ThermoFisher Scientific) and primers (Synbio Technologies) for 16S or 18S. The reaction volume was 20 µl and the mixture was incubated at 95 °C for 15 min.

Histopathological evaluation of lung tissue sections. Mouse lung sections were stained with H&E and PAS. The inflammatory score was calculated by quantifying the inflammatory cells stained with H&E in four randomly selected sections per mouse. A score of zero represented bronchi with no surrounding leukocyte infiltration, a score of one was assigned to those with few infiltrating leukocytes, and scores of two or three indicated that there were 1–2 or 3–5 layer(s) of perivascular and/or peribronchial leukocytes, respectively⁴¹. The proportion of epithelial cells that stained with PAS was determined for randomly selected bronchi²⁶. For each mouse, five randomly selected fields were assessed in PAS-stained cell calculation. Images were acquired using an Olympus BX53 microscope.

Flow cytometry. BALF cells were collected as described earlier. The total live BALF cell number was counted after Trypan blue (302643, Sigma-Aldrich) staining. The proportions of immune cells in the BALF cells were determined by flow cytometry¹⁰⁶. Briefly, the cells were blocked with mouse Fc block (130-092-575, Miltenyi Biotec) and stained with eBioscience fixable viability dye eFluor 780 (65-0865-14, ThermoFisher Scientific). For surface-marker detection, the cells were incubated with antibodies to CD3 (100204, BioLegend), CD4 (100408, BioLegend), SiglecF (155508, BioLegend), CD11b (101212, BioLegend), F4/80 (123116, BioLegend), Ly6G (127606, BioLegend), CD19 (152403, BioLegend), CD11c (117322, BioLegend) and MHC-II (107606, BioLegend) for 30 min at 4 °C. For GATA3 staining, the cells were treated with eBioscience transcription factor staining buffer set (00-5523-00, ThermoFisher Scientific) and stained with anti-GATA3 (653806, BioLegend). Flow cytometry was performed using a CytoFLEX flow cytometer (Beckman Coulter). Data were analysed using FlowJo 10.0 (Treestar). The percentages of F4/80⁺, CD19⁺, CD3⁺, CD3⁺CD4⁺GATA3⁺, CD11c⁺SiglecF⁺, CD11b⁺Ly6G⁺ and F480⁺CD11c⁺MHC-II⁺ cells in total live cells, determined by flow cytometry, were multiplied by the number of total live cells to quantify the macrophages, B cells, T cells, Th2, eosinophils, neutrophils and dendritic cells in the BALF.

Detection of LDH in BALF. The levels of LDH in the BALF supernatants were examined using a LDH cytotoxicity detection kit (8078, Sciencell) according to the manufacturer's instructions. The absorbance at 490 nm was recorded using a microplate reader (infinite M200 PRO).

Airway microvascular leakage assessment. Mice were injected with 1% Evans blue dye (E2129, Sigma-Aldrich; 100 mg kg⁻¹) via the tail vein 1 h before harvest. The mice were then anaesthetized and perfused with saline. After measuring the wet weight, the lung tissues were lysed with 1 M KOH at 37 °C for 24 h, followed by centrifugation. The concentration of Evans blue dye in the supernatant was measured at 620 nm using a microplate reader (infinite M200 PRO). The wet weight of the lung was divided by the concentration of Evans blue dye to quantify the exudation of Evans blue¹⁰⁷.

Protease activity assay. To determine whether there was DNase I contamination in the protease that we used, the protease activity of DNase I (89836, ThermoFisher Scientific) was determined using a Pierce protease assay kit (cat. no. 23263, ThermoFisher Scientific) according to the manufacturer's instructions¹⁰⁸. The optical density at 450 nm was recorded using a microplate reader (infinite M200 PRO).

Western blotting. RIPA buffer (p0013b, Beyotime) with proteinase inhibitor (78446, ThermoFisher Scientific) was used to lyse cells. To detect the translocation of PKC α and CRTCI, H146 cells (1×10^5 per well) subjected to the indicated treatments were stimulated with EETs (induced from 2×10^5 eosinophils) for 40 min and 2 h, respectively^{44,109,110}. Membrane, cytosolic and nuclear subcellular fractions were isolated using a Subcellular fractionation kit (78840, ThermoFisher Scientific). Proteins were analysed by SDS-PAGE and transferred to nitrocellulose membranes (3010040001, Roche). After blockade, the membranes were incubated at 4 °C overnight with antibodies to PAD4 (GTX113945, GeneTex), H3cit (ab5103, Abcam), CCDC25 (21209-1-AP, Proteintech), CRTCI (10441-1-AP, Proteintech), histone H1 (ab154111, Abcam), ILK (3862, Cell Signaling Technology), PKC α (2056, Cell Signaling Technology), Na-K-ATPase (96124, Cell Signaling Technology), GAPDH (2118S, Cell Signaling Technology) or tubulin (14652, Cell Signaling Technology), followed by incubation with horseradish peroxidase-linked antibody (7074S, Cell Signaling Technology). Immunoreactive bands were visualized using an iBright CL1000 system (ThermoFisher Scientific).

Quantitation of chromatin decondensation. Human eosinophils were cultured in plates pretreated with poly-L-lysine, incubated with or without DMSO or 50 μ M BB-Cl-amidine (HY-111347A, MCE) for 30 min and stimulated with 20 nM PMA at 37 °C for 2 h (refs. ^{36,77}). The samples were then sequentially incubated with AF555-anti-human/mouse ECP (bs-1754R-AF555, Bioss) and 5 μ M SYTO Green (S7573, ThermoFisher Scientific). Chromatin decondensation was quantified by calculating the average SYTO Green⁺ nucleus area of ECP⁺ cells using ImageJ (version 1.53c).

RT-qPCR. Mouse lung tissues were lysed using TissueLyser II (Qiagen). Total RNA was extracted from mouse lung tissues and H146 cells using TRIzol reagent¹¹¹. SYBR Green master mix kit (4309155, ThermoFisher Scientific) was used in the RT-qPCR procedures according to the manufacturer's instructions¹¹². The primer sequences (Synbio Technologies) are listed in Supplementary Table 2. Data were collected and analysed using a LightCycler 480 (Roche).

Cell lines. Human H146 cells (HTB-173) acquired from American Type Culture Collection were cultured in RPMI 1640 medium supplemented with 10% FBS. The authentication of H146 cells was performed by short tandem repeat profiling. The cells tested negative for mycoplasma.

Fluorescence-based assay for EYFP-CGRP expression. An EYFP-tagged CGRP plasmid was constructed by IGE Biotechnology Ltd^{113,114}. Briefly, genes encoding CGRP and EYFP were amplified using polymerase. A shuttle plasmid, pCDH-CMV-MCS-EF1-puro, was constructed to ligate genes encoding CGRP and EYFP, with EYFP being inserted at the 3' end of CGRP.

After transfection with EYFP-CGRP, H146 cells (1×10^5 per well) were seeded in 24-well plates with glass slides and treated with EETs, which were induced from 2×10^5 eosinophils with or without 1.5 U ml⁻¹ DNase I for 24 h at 37 °C (ref. ¹²). For neutralization of CCDC25, H146 cells were treated with EETs and 5 μ g ml⁻¹ anti-human CCDC25, as described in our previous study¹⁵, or IgG (PRABP01, Bio-Rad) concomitantly. Next, the cells were washed with PBS and stained with 0.5 μ M mCLING (membrane-binding fluorophore-cysteine-lysine-palmitoyl group)-488; 710006AT3, Synaptic Systems) for 10 min at 37 °C (ref. ¹¹⁵). The samples were fixed and imaged using a Leica TCS SP8 STED super-resolution microscope. The relative MFI of EYFP-CGRP in the H146 cells was quantified using Imaris 9.0 (Bitplane).

Isolation of eosinophil granule proteins. Eosinophil granule proteins were extracted from human eosinophils isolated from the peripheral blood of healthy donors^{52,53,116}. Briefly, human eosinophils were incubated in 0.25 M sucrose (V900116, Sigma-Aldrich) containing 250 U ml⁻¹ heparin (H6279, Sigma-Aldrich) and repeatedly ground for 20 min. The samples were sequentially centrifuged at 400g for 10 min and 10,000g for 20 min. The granules were resuspended with 10 mM HCl, processed with an ultrasonic cell crusher (SONICS) for 2 min and adjusted to pH 3.0. After centrifugation at 40,000g for 20 min, the granule proteins in the supernatants were enriched by ultrafiltration.

In vitro stimulation of H146 cells. For in vitro stimulation of H146 cells with EETs, H146 cells (1×10^5 per well) were treated with EETs, which were induced from 2×10^5 eosinophils in 500 μ l DMEM with or without 1.5 U ml⁻¹ DNase I at 37 °C for 24 h.

For in vitro stimulation of H146 cells with DNA, H146 cells (1×10^5 per well) were treated with or without 5 μ g ml⁻¹ 8-OHdG-DNA (HY-W011540, MCE) or 5 μ g ml⁻¹ EET-DNA (isolated as described earlier) at 37 °C for 24 h. The H146 cells were concomitantly incubated with 7.5 μ g ml⁻¹ eosinophil granule proteins,

recombinant protein EPX (MBS653827, MyBioSource), MBP (abx067909, Abnova Ltd), ECP (230-20270-200, RayBiotech), eosinophil-derived neurotoxin (MBS967723, MyBioSource) or galectin-10 (NBPI-51096, Novus) in some experiments.

To induce CRTCI nuclear translocation, H146 cells were stimulated with 25 mM forskolin (HY-15371, MCE) and 30 mM KCl 1 h before EET treatment⁴⁴.

Ca²⁺ influx assay using Fura-2. The influx of cytosolic Ca²⁺ was analysed using a Fura-2 ratiometric assay^{117,118}. Briefly, H146 cells (5×10^3 per well in 96-well plates) pretreated with Fura-2/AM (F1201, Invitrogen) were stimulated with EETs (induced from 1×10^4 eosinophils) in the presence or absence of 1.5 U ml⁻¹ DNase I. The cytosolic Ca²⁺ concentration was measured by the F340/F380 ratio (evoked at 340 nm and 380 nm; emitted at 510 nm) using a microplate reader (Sunrise).

Preparation of lung slices and staining of PNECs. Live lung slices (200–300 μ m thick) were prepared from the lungs of female C57BL/6 mice (2–3 weeks old) using a vibratome (Leica VT1200s)^{119,120}. For staining of PNECs, the lung slices were incubated with 4 μ M 4-Di-2-ASP (D3418, Sigma-Aldrich) in DMEM/F-12 medium (11330032, ThermoFisher Scientific) for 10 min at 37 °C. The slices were then washed and kept in DMEM/F-12 in an incubator at 37 °C.

Loading of Ca²⁺ indicator. Mouse lung slices were incubated with 10 μ M Fluo-4 AM (F14201, Invitrogen) and 0.02% Pluronic F-127 (P2443, Sigma-Aldrich) for 30 min at 37 °C (ref. ¹¹⁹). After washing with saline, the slices were placed in confocal dishes, immobilized and treated with 50 μ l of EETs in the presence or absence of 1.5 U ml⁻¹ DNase I. Images were acquired using an LSM 880 microscope with Airyscan (Carl Zeiss). The fluorescence intensity of Fluo-4 in PNECs was measured using Imaris 9.0.

siRNA-mediated silencing. To induce gene silencing in H146 cells, H146 cells (2×10^5 cells ml⁻¹) were cultured in six-well plates and transfected with 25 nM specific siRNA duplexes (generated by Hongxun Biotechnologies) with Lipofectamine 3000 (L3000-015, ThermoFisher Scientific) according to the manufacturer's instructions¹²¹. The oligonucleotide sequences of the siRNAs are listed in Supplementary Table 3. Two days after transfection, the H146 cells were harvested for immunoblots or treated with EETs.

CRISPR-Cas9-mediated gene knockout. CCDC25 was knocked out of H146 cells using the CRISPR-Cas9 system with the Cas9 lentivirus and gRNA lentivirus (GenePharma), followed by 2.5 μ g ml⁻¹ puromycin selection for 2 weeks¹⁵. The sequences targeting CCDC25 (gRNA) were 5'-GTCTGGGAACGCTCGACTT-3' and 5'-GAATGCTATTGGCCTTACA-3'.

Lentiviral vector-mediated silencing. Lentiviral vector LV3-pGLV-H1-GFP/Puro (GenePharma) was used for knockdown experiments. The oligonucleotide sequences of shRNAs against human CRTCI, ILK and PKC α are listed in Supplementary Table 4. H146 cells were transduced with 5×10^7 lentiviral particles containing shRNAs (multiplicity of infection of 10–20) with 8 μ g ml⁻¹ polybrene (TR-1003, Sigma-Aldrich).

Statistics and reproducibility. Data are presented as the mean $\pm$ s.d. For the comparison of two groups, data were analysed using a two-tailed unpaired Students' *t*-test with a 95% confidence interval. For the comparison of three or more groups, data were analysed using a one-way or two-way ANOVA, as indicated in the figure legends. *P* values of less than 0.05 were considered significant. Linear regression was used to identify correlations between two variables. Spearman's order (for discontinuous variables) and Pearson's (for continuous variables) correlation analyses were performed to determine the relationship between two different factors. Statistical analyses were performed using GraphPad Prism 7.0.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Mass spectrometry data for EET-binding proteins have been deposited in ProteomeXchange with the identifier PXD027951 (<https://www.ebi.ac.uk/pride/>)¹²². Data of human CCDC25 RNA expression was derived from the FANTOM5 dataset included in the Human Protein Atlas database (<https://www.proteinatlas.org/ENSG00000147419-CCDC25/tissue>)³⁷. All other data supporting the findings of this study are available from the corresponding author on reasonable request. Source data are provided with this paper.

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Author contributions

S.S. conceived the study, designed the experiments and wrote the manuscript. S.J. and Y.H. provided the clinical samples. Y.L., Y.H., Jiang Li, J.H. and L.Z. performed the experiments, analysed the data and wrote the manuscript. J.F., Jiaqian Li., Q.X., Q.Z. and L.H. performed the experiments and analysed the data.

Competing interests

The authors declare no competing interests.

Additional information

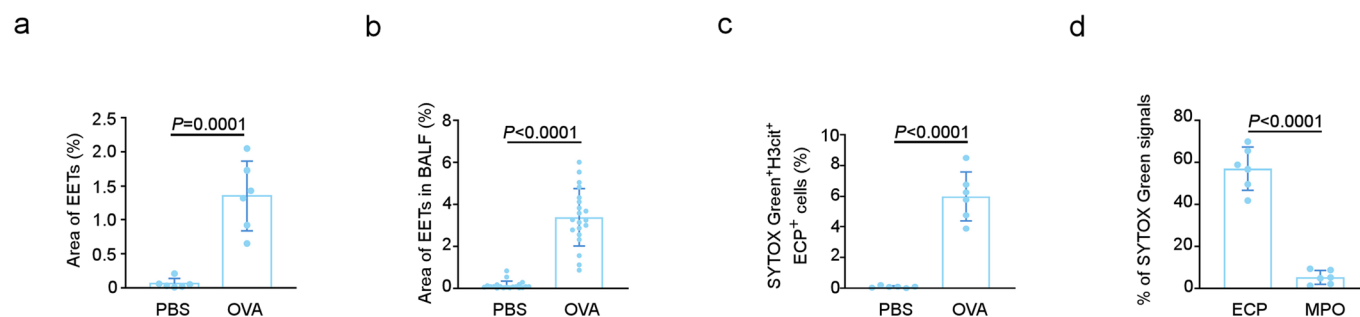
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Correspondence and requests for materials should be addressed to Shanping Jiang or Shicheng Su.

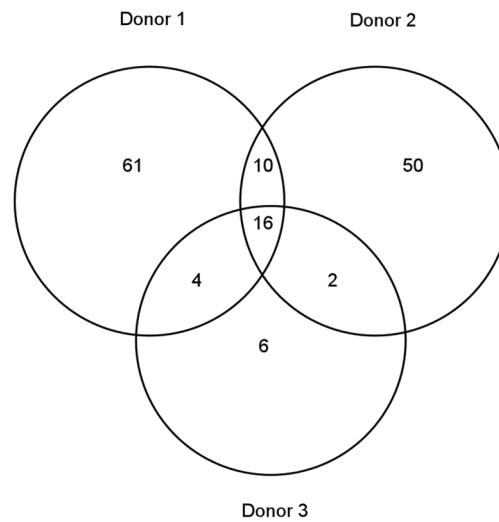
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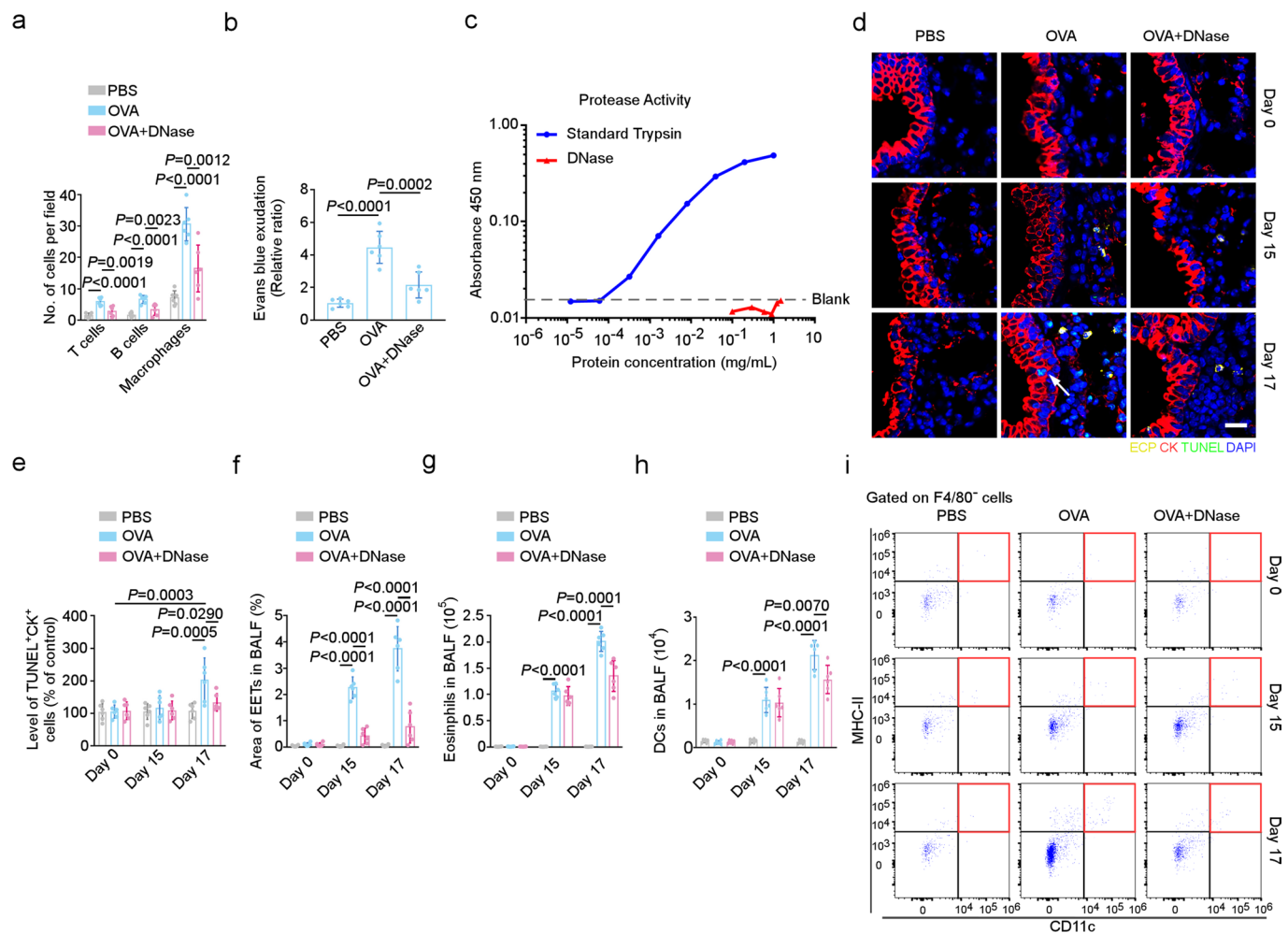


Extended Data Fig. 1 | EETs are correlated with asthma progression. a, Quantification of Fig. 1d. **b**, Levels of EETs were evaluated by the SYTOX Green, DAPI and ECP staining in BALF of mice immunized without or with OVA. **c**, Quantification of Fig. 1f. **d**, Quantification of SYTOX Green signals in ECP⁺ or MPO⁺ cells in BALF of mice immunized with OVA in Fig. 1e. **a–d**, Mean $\pm$ s.d.; $n = 6$ mice per group, representative of two independent experiments (**a**, **c**, **d**); Data are pooled from 3 independent experiments with 6–8 mice each (**b**); compared by two-tailed unpaired Student's *t*-test.

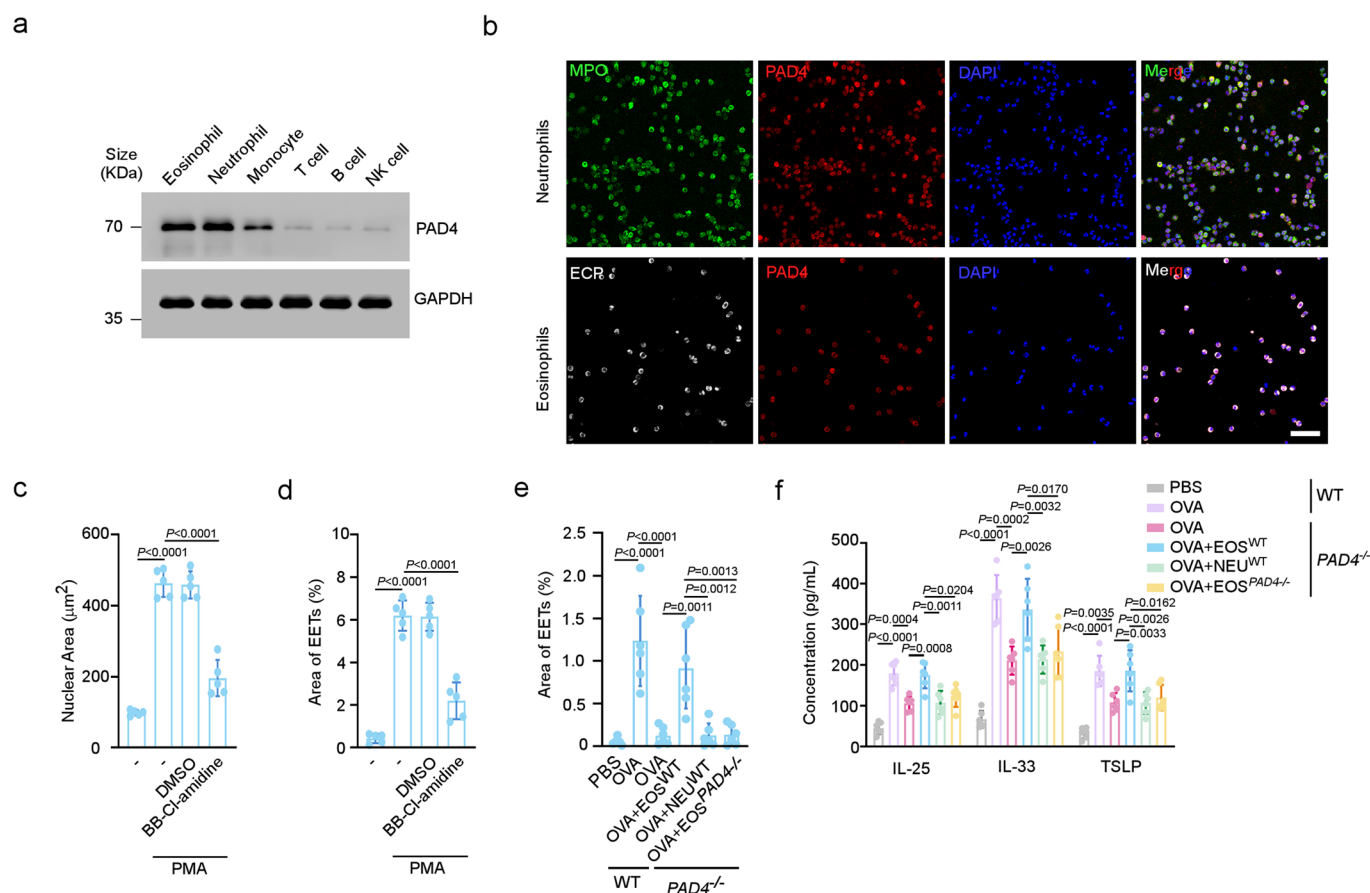
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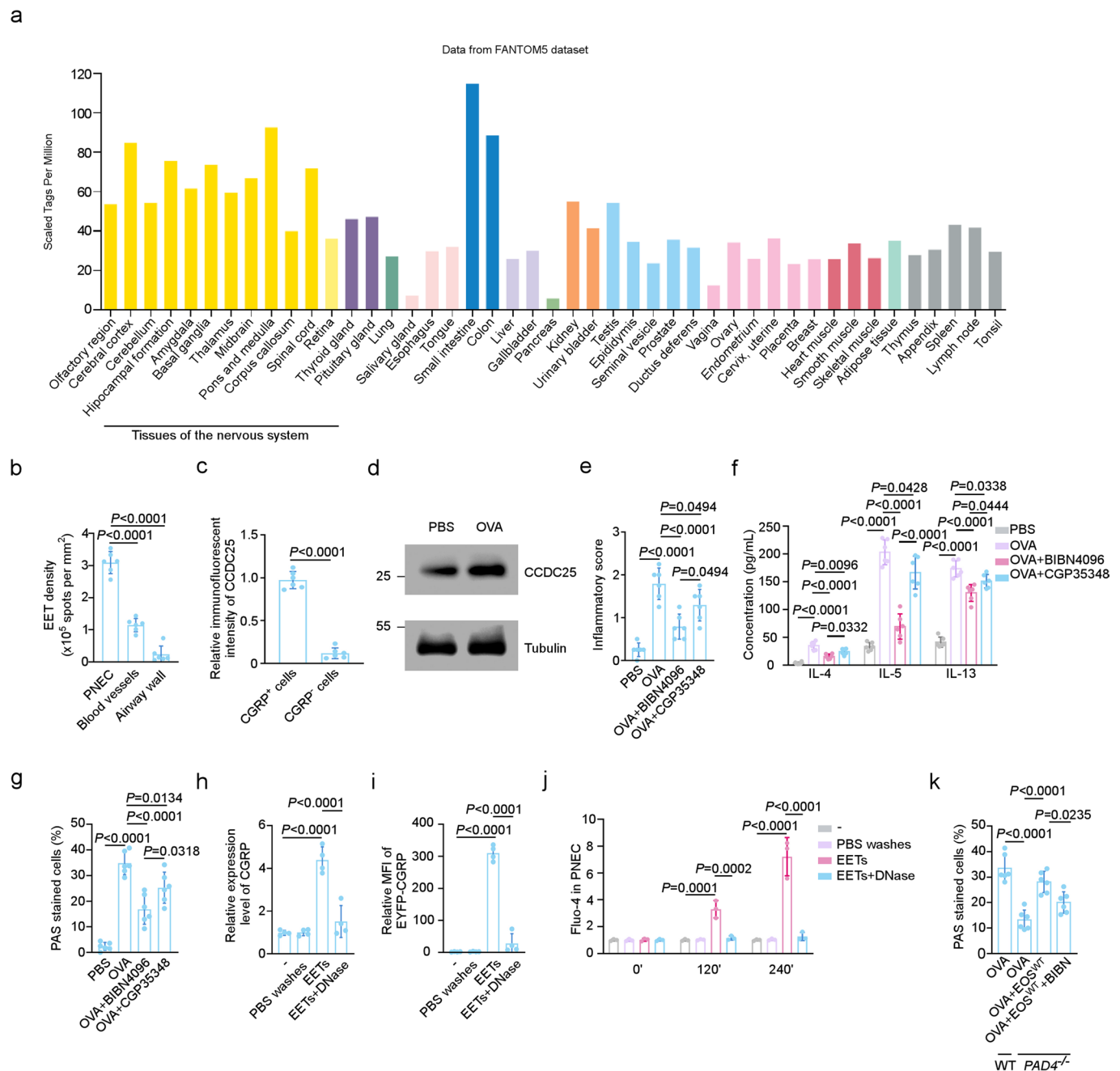
Extended Data Fig. 2 | Generation and components of EETs. a, Venn diagram of overlapped proteins from three donors in Fig. 2g. The numbers of overlapped and unique proteins are shown in the corresponding area.



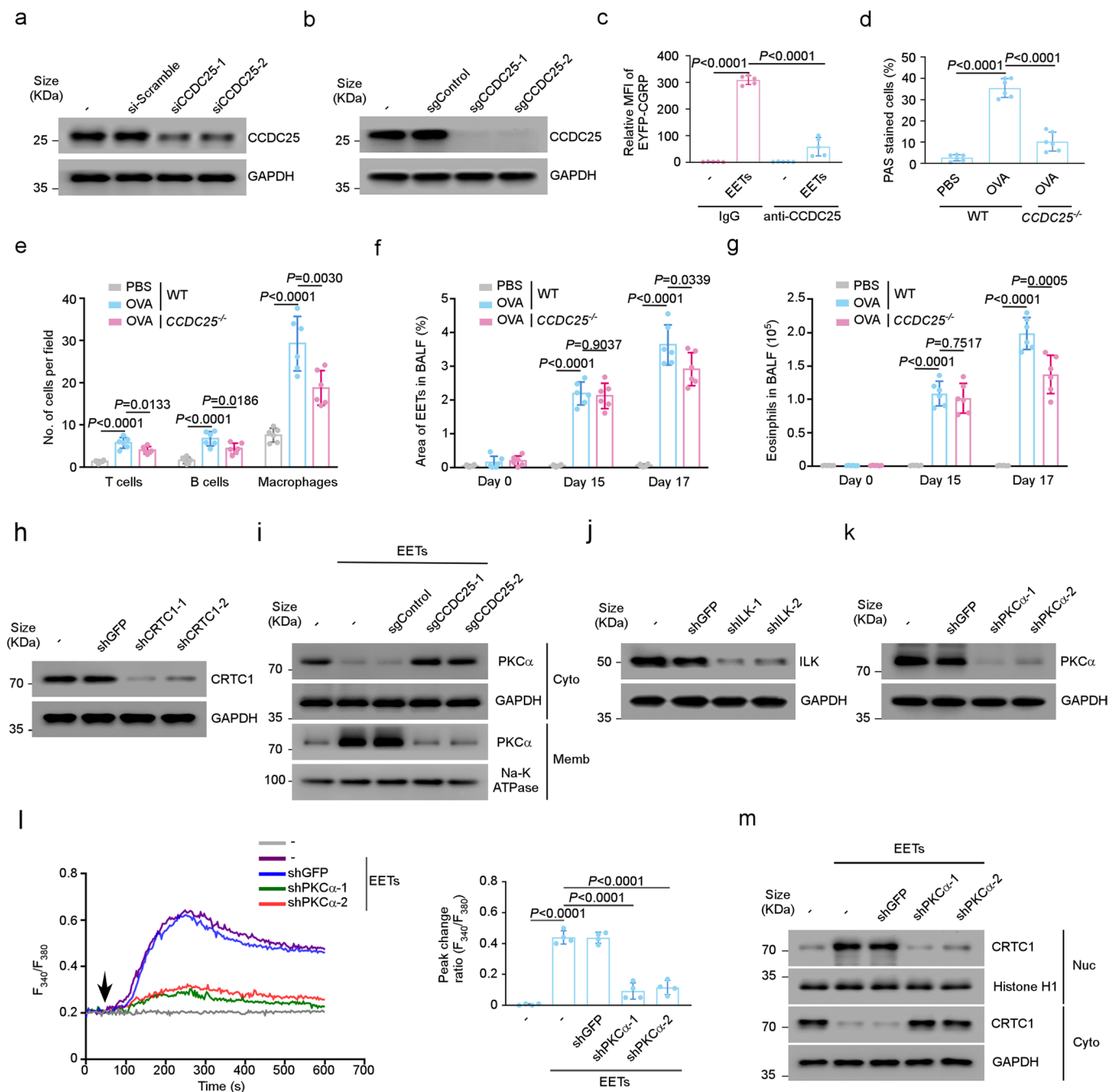
Extended Data Fig. 3 | DNase treatment ameliorates asthma in mice. **a**, Quantification of Fig. 3f. **b**, Quantification of Fig. 3h. **c**, Protease activity of DNase I at indicated concentrations was examined by the protease assay. Trypsin was used as a positive control ($n=3$). **d-i**, Wild-type mice were sensitized and challenged with PBS or OVA. OVA-sensitized mice were treated intratracheally with DNase before OVA challenge. **d**, Representative immunofluorescence images of cytokeratin (CK), TUNEL and DAPI staining in sections of lungs harvested at baseline (day 0) and 15 (day 15) and 17 (day 17) days after the first OVA sensitization. The arrow indicates CK⁺ TUNEL⁺ cells. Scale bar, 20 μ m. **e**, Quantification of **d**. **f**, The area (%) of ECP⁺ SYTOX Green⁺ EETs in BALF smears was quantified at baseline (day 0) and 15 (day 15) and 17 (day 17) days after the first OVA sensitization. **g**, Eosinophils in the BALF at baseline (day 0) and 15 (day 15) and 17 (day 17) days after the first OVA sensitization were evaluated by flow cytometric analysis. Quantification are shown. **h**, **i**, F4/80⁺ MHC-II⁺CD11c⁺ dendritic cells (DCs) in the BALF at baseline (day 0) and 15 (day 15) and 17 (day 17) days after the first OVA sensitization were evaluated by flow cytometric analysis. Quantification (**h**) and representative plots (**i**) are shown. **a**, **b**, **e-h**, Mean $\pm$ s.d.; $n=6$ mice per group, representative of two independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test (**a**, **b**, **f-h**) and two-way ANOVA with Sidak's multiple comparisons test (**e**).



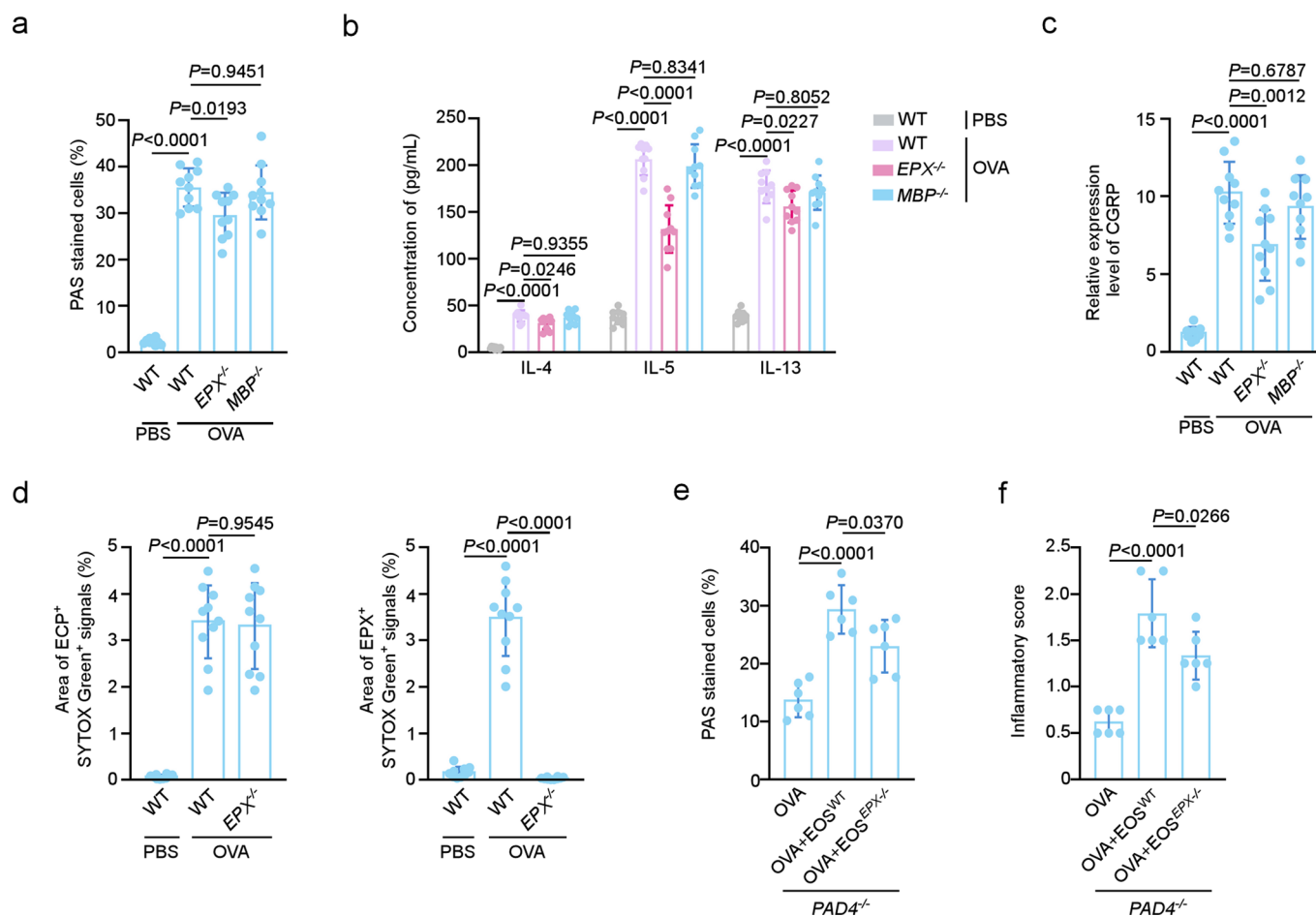
Extended Data Fig. 4 | EETs amplify allergic asthma responses. **a**, Representative immunoblots of PAD4 and GAPDH in human eosinophils, neutrophils, monocytes, T cell, B cells and NK cells isolated from peripheral blood of healthy donors ($n=3$). **b**, Representative immunofluorescence images of MPO, ECP, PAD4 and DAPI staining in human neutrophils and eosinophils isolated from peripheral blood of healthy donors. Scale bar, 50 μm . **c**, Quantification of Fig. 4b. **d**, Quantification of Fig. 4c. **e, f**, Wild-type and $PAD4^{-/-}$ mice were sensitized and challenged with PBS or OVA. $PAD4^{-/-}$ mice were adoptively transferred with 2×10^5 eosinophils or neutrophils isolated from wild-type or $PAD4^{-/-}$ mice. **e**, Quantification of Fig. 4e. **f**, Levels of IL-25, IL-33 and TSLP in BALF were evaluated by ELISA. **c-f**, Mean $\pm$ s.d.; $n=5$ independent experiments (**c, d**); $n=6$ mice per group, representative of two independent experiments (**e, f**); compared by one-way ANOVA with Tukey's multiple comparisons test.



Extended Data Fig. 5 | EETs aggravate allergic inflammation by activating PNECs. **a**, Expression profiles of CCDC25 in different tissues from the FANTOM5 dataset. **b**, ECP⁺ H3cit⁺ DAPI⁺ spots (0.1- μ m-diameter) in lung sections of OVA-immunized mice were defined as EETs as shown in Fig. 5a. The density of EETs (number of spots per mm²) within 50 μ m of PNECs, randomly selected blood vessels and airway walls (from epithelium to serosa, excluding the cartilage and 50- μ m area surrounding PNECs) was quantified. **c**, CCDC25 expression level of CGRP⁺ and CGRP⁻ cells in mouse lung sections was quantified by immunofluorescence as shown in Fig. 5b. **d**, Representative immunoblots of CCDC25 of lung homogenate from C57BL/6 mice challenged with PBS or OVA ($n=3$). **e-g**, C57BL/6 mice immunized with OVA were treated without or with BIBN4096 or CGP35348. **e**, The inflammatory score quantified from H&E staining of lung sections. **f**, Levels of indicated type 2 cytokines in BALF evaluated by ELISA. **g**, Percentages of PAS-stained goblet cells in total epithelial cells in lung sections. **h**, The relative expression level of CGRP in H146 cells with indicated treatment was measured by qRT-PCR. **i**, Quantification of Fig. 5e. **j**, The relative level of intracellular Fluor-4 immunofluorescence intensity in live lung slices with indicated treatment was quantified at indicated time point. **k**, Quantification of Fig. 5h. **b, c, e-k**, Mean $\pm$ s.d.; $n=6$ mice per group, representative of two independent experiments (**b, c, e-g, k**); $n=4$ (**h, i**) or 3 (**j**) independent experiments; compared by two-tailed unpaired Student's *t*-test (**c**) and one-way ANOVA with Tukey's multiple comparisons test (**b, e-k**).



Extended Data Fig. 6 | EETs activate PNECs via CCDC25. **a**, Representative immunoblots for CCDC25 in H146 cells with indicated treatment (n=3). **b**, Representative immunoblots of CCDC25 in H146 cells that were transduced with sgControl or sgRNAs against CCDC25 (n=3). **c**, Quantification of Fig. 6c. **d-g**, Wild-type and CCDC25^{-/-} mice were immunized with OVA. **d**, Quantification of Fig. 6e. **e**, Quantification of indicated cells in immunofluorescence images of lung sections. **f**, The area (%) of EETs in BALF smear at baseline (day 0) and 15 (day 15) and 17 (day 17) days after the first OVA sensitization was quantified by immunofluorescence staining. **g**, The number of eosinophils in the BALF at indicated time point was quantified by flow cytometry. **h**, CRTC1 was silenced by shRNA in H146 cells. The efficiency was evaluated by western blotting (n=3). **i**, Representative immunoblots of PKCα levels in membrane (Memb) and cytoplasmic (Cyto) fractions of H146 cells with indicated treatments (n=3). **j**, ILK (j) and PKCα (k) were silenced by shRNAs in H146 cells. The efficiency was evaluated by western blotting (n=3). **l**, H146 cells transduced with shRNAs against PKCα were treated with EETs. **l**, Cytosolic Ca²⁺ mobilization in H146 cells was examined using Fura-2/AM ratiometric fluorescence measurements. Fluorescence curves (left) and peak responses (right) are shown. Arrow indicates the time of stimulation. **m**, Representative immunoblots of CRTC1 in nuclear (Nuc) and cytoplasmic (Cyto) fractions of H146 cells (n=3). **c-g**, **l**, Mean ± s.d.; n=6 mice per group, representative of two independent experiments (**d-g**); n=5 (**c**) or 4 (**l**) independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test (**d-g**, **l**) and two-way ANOVA with Sidak's multiple comparisons test (**c**).



Extended Data Fig. 7 | EPX potentiates the interaction between EETs and PNECs. a-c, Wild-type, EPX^{-/-} and MBP^{-/-} mice were sensitized and challenged with PBS or OVA. **a,** Quantification of Fig. 7d. **b,** Levels of indicated type 2 cytokines in BALF were evaluated by ELISA. **c,** The relative expression levels of CGRP in lung homogenates were examined by qRT-PCR. **d,** The area (%) of ECP⁺ SYTOX Green⁺ and EPX⁺ SYTOX Green⁺ signals in BALF smears as shown in Fig. 7f. **e-f,** PAD4^{-/-} mice were adoptively transferred with eosinophils isolated from wild-type mice or EPX^{-/-} mice. **e,** Percentages of PAS-stained cells in total epithelial cells in lung sections. **f,** The inflammatory score quantified from the H&E staining of lung sections. **a-f,** Mean $\pm$ s.d.; $n=6$ (**e, f**) or 10 (**a-d**) mice per group, representative of two independent experiments; compared by one-way ANOVA with Tukey's multiple comparisons test.

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Data collection

Images were acquired with Olympus BX53 (Olympus), Laser Scanning Confocal Microscope 800 (Zeiss), Laser Scanning Confocal Microscope 880 (Zeiss), TCS SP8 STED super-resolution microscope (Leica) and FEI QUANTA 200 scanning electron microscope. CytoFLEX Flow Cytometer S (Beckman Coulter) was used for flow cytometry. LightCycler 480 instrument (Roche) was used for qRT-PCR. Microplate readers used for this research include Microplate reader (infinite M200 PRO) and Microplate Reader (Sunrise). EET-binding protein analysis was performed by ThermoFisher Scientific Q Exactive mass spectrometer.

Data analysis

Flow cytometry analysis was performed by FlowJo 10.0 (Treestar). Confocal images were analyzed by ZEN 2.3 (blue edition) and quantified by ImageJ (version 1.53c) or Imaris 9.0 with coloc function (Bitplane). Statistic analysis was done with Prism 7.

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- A list of figures that have associated raw data
- A description of any restrictions on data availability

All data involved in current study are included in article or its supplementary information and source data files. Mass spectrometry data for EET-binding proteins have been deposited in ProteomeXchange with the identifier PXD027951 (<https://www.ebi.ac.uk/pride/>). Uniprot Database (<https://www.uniprot.org/>) was used for analysis of mass spectrometry data. Data of human CCDC25 RNA expression was derived from the FANTOM5 dataset included in the Human Protein Atlas database (<https://www.proteinatlas.org/ENSG00000147419-CCDC25/tissue>). All other data supporting the findings of this study are available from the corresponding author on reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes of asthma patients and animal experiments were estimated on the basis of similar research on asthma (Toussaint et al, Nature Medicine, 2017; Simpson et al, Nature Immunology, 2014). The number of experimental subjects was indicated in figure legends.
Data exclusions	No data were excluded.
Replication	In vitro studies were replicated at least 3 times as indicated in figure legends. Animal studies were replicated in at least 2 independent experiments with 6-10 mice per group. All replications were successful.
Randomization	All the mice were randomly allocated into the different experimental groups. Randomization was implemented by label each mouse with an ID (for example, from 1 to 12 if there are two groups with six mice per group). A random number was generated for each ID using the Microsoft Excel's function "RAND()". The no. of treatment groups (for example, 1,1,1,1,1,1 and 2,2,2,2,2,2 for two groups with six mice per group) was generated. Then the no. of treatment group was sorted based on the values of the random number. Thus, each ID was allocated to different treatment groups randomly.
Blinding	Investigators were blinded during data collection and analysis including quantification of immunohistochemical and immunofluorescent images.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Antibodies used for flow cytometry:
anti-mouse/human histone H3 (citrulline 2,8,17) antibody (ab5103, Abcam; 1:30), anti-mouse ECP (Cat# MBS2090412, MyBioSource; 1:50), Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (A21428, ThermoFisher Scientific, 1:500),

Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (A21235, ThermoFisher Scientific; 1:500), FITC anti-CD3 (Cat# 100204, Biolegend; 1:20), PE anti-CD4 (Cat# 100408, Biolegend; 1:20), APC anti-SiglecF (Cat# 155508, Biolegend; 1:20), APC anti-CD11b (Cat# 101212, Biolegend; 1:20), APC anti-F4/80 (Cat# 123116, Biolegend; 1:20), FITC anti-Ly6G (Cat# 127606, Biolegend; 1:20), FITC anti-CD19 (Cat# 152403, Biolegend; 1:20), Pacific Blue anti-CD11c (Cat# 117322, Biolegend; 1:20), FITC anti-MHC-II (Cat# 107606, Biolegend; 1:20), APC anti-GATA3 (Cat# 653806, Biolegend; 1:20).

Antibodies for western blotting:

PAD4 (Cat# GTX113945, GeneTex; 1:1000), H3cit (Cat# ab5103, Abcam; 1:1000), CCDC25 (Cat# 21209-1-AP, Proteintech; 1:1000), CRTC1 (Cat# 10441-1-AP, Proteintech; 1:1000), Histone H1 (Cat# ab154111, Abcam; 1:1500), ILK (Cat# 3862, Cell Signaling Technology; 1:1000), PKC α (Cat# 2056, Cell Signaling Technology; 1:1000), Na-K-ATPase (Cat# 96124, Cell Signaling Technology; 1:1000), GAPDH (Cat# 2118S, Cell Signaling Technology; 1:2000), Tubulin (Cat# 14652, Cell Signaling Technology; 1:3000), anti-rabbit IgG HRP-linked antibody (Cat# 7074S, Cell Signaling Technology; 1:3000)

Antibodies for immunofluorescence:

rabbit anti-human/mouse ECP antibody conjugated with Alexa Fluor 555 (Cat# bs-1754R-AF555, Bioss; 1:100), anti-human ECP (Cat# MA5-34931, ThermoFisher Scientific; 1:50), anti-mouse ECP (Cat# MBS2090412, MyBioSource; 1:50), anti-human/mouse H3cit (Cat# ab5103, Abcam; 1:50), anti-mouse CD3 (Cat# MAB4841, R&D; 1:100), anti-mouse CD19 (Cat# 3574S, Cell Signaling Technology; 1:50), anti-mouse F4/80 (Cat# 30325, Cell Signaling Technology; 1:50), anti-mouse CGRP (Cat# PA1-85250, ThermoFisher Scientific; 1:100), anti-human/mouse CCDC25 (Cat# 21209-1-AP, Proteintech; 1:100), anti-mouse pan cytokeratin (Cat# ab9377, Abcam; 1:100), anti-human/mouse MPO (Cat# AF3667, R&D Systems; 1:100), anti-mouse EPX (Cat# orb382913, Biorbyt; 1:100), anti-human/mouse CRTC1 (Cat# orb447387, Biorbyt; 1:100), anti-human/mouse PAD4 (Cat# GTX113945, GeneTex; 1:100), Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A21202, ThermoFisher Scientific; 1:400), Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A31570, ThermoFisher Scientific; 1:400), Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (Cat# A31571, ThermoFisher Scientific; 1:400), Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A21206, ThermoFisher Scientific; 1:400), Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A31572, ThermoFisher Scientific; 1:400), Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (Cat# A31573, ThermoFisher Scientific; 1:400), Donkey anti-Rat IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# 21208, ThermoFisher Scientific; 1:400), Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A11055, ThermoFisher Scientific; 1:400), Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A21432, ThermoFisher Scientific; 1:400), Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 594 (Cat# A11058, ThermoFisher Scientific; 1:400)

For in vivo experiments:

anti-IL-4 (Cat# 14-7041-85, ThermoFisher Scientific; 0.125 mg per mouse), anti-IL-5 (Cat# 14-7052-81, ThermoFisher Scientific; 0.125 mg per mouse), anti-IL-13 (Cat# 16-7135-85, ThermoFisher Scientific; 0.075 mg per mouse), anti-IL-33 (Cat# AF-3626, R&D Systems; 0.075 mg per mouse), anti-IL-25 (Cat# MAB13992, R&D Systems; 0.5 mg per mouse), anti-TSLP (Cat# MAB555, R&D Systems; 0.025 mg per mouse), IgG (Cat# MAB006R, R&D Systems; 0.025 mg per mouse).

Validation

All antibodies used in this study were obtained from commercial sources and validated according to manufacturers' instruction.

Antibodies used for flow cytometry:

Rabbit anti-mouse/human histone H3 (citrulline 2,8,17) antibody (ab5103, Abcam), <https://www.abcam.cn/histone-h3-citrulline-r2--r8--r17-antibody-ab5103.html>

Mouse anti-mouse ECP (Cat# MBS2090412, clone: D4, MyBioSource), <https://www.mybiosource.com/monoclonal-mouse-antibody/ribonuclease-a3-rnase3/2090412>

Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (A21428, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Goat-anti-Rabbit-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21428>

Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (A21235, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Goat-anti-Mouse-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21235>

FITC anti-mouse CD3 (Cat# 100204, clone: 17A2, Biolegend), <https://www.biolegend.com/en-us/products/fits-anti-mouse-cd3-antibody-45>

PE anti-mouse CD4 (Cat# 100408, clone: GK1.5, Biolegend), <https://www.biolegend.com/en-us/products/pe-anti-mouse-cd4-antibody-250>

APC anti-mouse SiglecF (Cat# 155508, clone: S17007L; Biolegend), <https://www.biolegend.com/en-us/products/apc-anti-mouse-cd170-siglec-f-antibody-16373>

APC anti-mouse CD11b (Cat# 101212, clone: M1/70, Biolegend), <https://www.biolegend.com/en-us/products/apc-anti-mouse-human-cd11b-antibody-345>

APC anti-mouse F4/80 (Cat# 123116, clone: BM8, Biolegend), <https://www.biolegend.com/en-us/products/apc-anti-mouse-f4-80-antibody-4071>

FITC anti-mouse Ly6G (Cat# 127606, clone: 1A8, Biolegend), <https://www.biolegend.com/en-us/products/fits-anti-mouse-ly-6g-antibody-4775>

FITC anti-mouse CD19 (Cat# 152403, clone: 1D3/CD19, Biolegend), <https://www.biolegend.com/en-us/products/fits-anti-mouse-cd19-antibody-13615>

Pacific Blue anti-mouse CD11c (Cat# 117322, clone: N418, Biolegend), <https://www.biolegend.com/en-us/products/pacific-blue-anti-mouse-cd11c-antibody-3864>

FITC anti-mouse MHC-II (Cat# 107606, clone: M5/114.15.2, Biolegend), <https://www.biolegend.com/en-us/products/fits-anti-mouse-i-a-i-e-antibody-366>

APC anti-GATA3 (Cat# 653806, clone: 16E10A23, Biolegend), <https://www.biolegend.com/en-us/products/apc-anti-gata3-antibody-9104>

Antibodies for western blotting:

Rabbit anti-human PAD4 (Cat# GTX113945, GeneTex), <https://www.genetex.cn/Product/Detail/PAD4-antibody/GTX113945>

Rabbit anti-mouse/human histone H3 (citrulline 2,8,17) antibody (ab5103, Abcam), <https://www.abcam.cn/histone-h3-citrulline-r2--r8--r17-antibody-ab5103.html>

Rabbit anti mouse CCDC25 (Cat# 21209-1-AP, Proteintech), <https://www.ptgcn.com/products/CCDC25-Antibody-21209-1-AP.htm>
 Rabbit anti human CRTCl (Cat# 10441-1-AP, Proteintech), <https://www.ptgcn.com/products/CRTCl-Antibody-10441-1-AP.htm>
 Rabbit anti human Histone H1 (Cat# ab154111, Abcam), <https://www.abcam.cn/histone-h10-antibody-ab154111.html>
 Rabbit anti-human ILK (Cat# 3862, Cell Signaling Technology), https://www.cellsignal.cn/products/primary-antibodies/ilk1-antibody/3862?site-search-type=Products&N=4294956287&Ntt=3862&fromPage=plp&_requestid=2361382
 Rabbit anti-human PKC α (Cat# 2056, Cell Signaling Technology), https://www.cellsignal.cn/products/primary-antibodies/pkca-antibody/2056?site-search-type=Products&N=4294956287&Ntt=2056&fromPage=plp&_requestid=2361531
 Rabbit anti-human Na-K-ATPase (Cat# 96124, clone: D4Y7E, Cell Signaling Technology), https://www.cellsignal.cn/products/antibody-conjugates/na-k-atpase-a1-d4y7e-rabbit-mab-hrp-conjugate/96124?site-search-type=Products&N=4294956287&Ntt=96124&fromPage=plp&_requestid=2361631
 Rabbit anti-human GAPDH (Cat# 2118S, clone: 14C10, Cell Signaling Technology), https://www.cellsignal.cn/products/primary-antibodies/gapdh-14c10-rabbit-mab/2118?site-search-type=Products&N=4294956287&Ntt=2118s&fromPage=plp&_requestid=2361709
 Rabbit anti-mouse Tubulin (Cat# 14652, Cell Signaling Technology), https://www.cellsignal.cn/products/antibody-conjugates/acetyl-a-tubulin-lys40-d20g3-xp-rabbit-mab-hrp-conjugate/14652?site-search-type=Products&N=4294956287&Ntt=14652&fromPage=plp&_requestid=2361871
 Anti-rabbit IgG HRP-linked antibody (Cat# 7074S, Cell Signaling Technology), https://www.cellsignal.cn/products/secondary-antibodies/anti-rabbit-igg-hrp-linked-antibody/7074?site-search-type=Products&N=4294956287&Ntt=7074s&fromPage=plp&_requestid=2361932

Antibodies for immunofluorescence:

Rabbit anti-human/mouse ECP antibody conjugated with Alexa Fluor 555 (Cat# bs-1754R-AF555, Bioss; 1:100), http://www.bioss.com.cn/prolook_03_biaoj.asp?pro2a=201208160816361620&pro33=217
 Mouse anti-human ECP (Cat# MA5-34931, clone: 5G4, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/RNASE3-Antibody-clone-5G4-Monoclonal/MA5-34931>
 Mouse anti-mouse ECP (Cat# MBS2090412, clone: D4, MyBioSource), <https://www.mybiosource.com/monoclonal-mouse-antibody/ribonuclease-a3-rnase3/2090412>
 Rabbit anti-mouse/human histone H3 (citrulline 2,8,17) antibody (ab5103, Abcam), <https://www.abcam.cn/histone-h3-citrulline-r2--r8--r17-antibody-ab5103.html>
 Rat anti-mouse CD3 (Cat# MAB4841, clone: 17A2, R&D), https://www.rndsystems.com/cn/products/mouse-cd3-antibody-17a2_mab4841
 Rabbit anti-mouse CD19 (Cat# 3574S, Cell Signaling Technology), https://www.cellsignal.cn/products/primary-antibodies/cd19-antibody/3574?site-search-type=Products&N=4294956287&Ntt=3574s&fromPage=plp&_requestid=2362346
 Rabbit anti-mouse F4/80 (Cat# 30325, clone: D4C8V, Cell Signaling Technology), https://www.cellsignal.cn/products/primary-antibodies/f4-80-d4c8v-xp-rabbit-mab/30325?site-search-type=Products&N=4294956287&Ntt=30325%E3%80%81&fromPage=plp&_requestid=2362407
 Goat anti-mouse CGRP (Cat# PA1-85250, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/CGRP-Antibody-Polyclonal/PA1-85250>
 Rabbit anti mouse CCDC25 (Cat# 21209-1-AP, Proteintech), <https://www.ptgcn.com/products/CCDC25-Antibody-21209-1-AP.htm>
 Rabbit anti-mouse pan cytokeratin (Cat# ab9377, Abcam), <https://www.abcam.cn/wide-spectrum-cytokeratin-antibody-ab9377.html>
 Goat anti-human/mouse MPO (Cat# AF3667, R&D Systems), https://www.rndsystems.com/cn/products/human-mouse-myeloperoxidase-mpo-antibody_af3667
 Rabbit anti-mouse EPX (Cat# orb382913, Biorbyt), <https://www.biorbyt.com/solr/index/?q=orb382913>
 Rabbit anti human CRTCl (Cat# orb447387, Biorbyt), <https://www.biorbyt.com/crtcl-antibody-biotin-orb447387.html>
 Rabbit anti-human PAD4 (Cat# GTX113945, GeneTex), <https://www.genetex.cn/Product/Detail/PAD4-antibody/GTX113945>
 Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A21202, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21202>
 Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A31570, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-31570>
 Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (Cat# A31571, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-31571>
 Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A21206, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21206>
 Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A31572, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-31572>
 Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 647 (Cat# A31573, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-31573>
 Donkey anti-Rat IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# 21208, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Rat-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21208>
 Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 488 (Cat# A11055, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11055>
 Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 555 (Cat# A21432, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21432>
 Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 594 (Cat# A11058, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11058>

For in vivo experiments:

anti-IL-4 (Cat# 14-7041-85, clone: 11B11, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/IL-4-Antibody-clone-11B11-Monoclonal/14-7041-85>

anti-IL-5 (Cat# 14-7052-81, clone: TRFK5 ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/IL-5-Antibody-clone-TRFK5-Monoclonal/14-7052-81>
 anti-IL-13 (Cat# 16-7135-85, clone: eBio1316H, ThermoFisher Scientific), <https://www.thermofisher.cn/cn/en/antibody/product/IL-13-Antibody-clone-eBio1316H-Monoclonal/16-7135-85>
 anti-IL-33 (Cat# AF-3626, R&D Systems), https://www.rndsystems.com/cn/products/mouse-il-33-antibody_af3626
 anti-IL-25 (Cat# MAB13992, clone: 881014, R&D Systems), https://www.rndsystems.com/cn/products/mouse-il-17e-il-25-antibody-881014_mab13992
 anti-TSLP (Cat# MAB555, clone: 152614, R&D Systems), https://www.rndsystems.com/cn/products/mouse-tslp-antibody-152614_mab555
 IgG (Cat# MAB006R, R&D Systems), https://www.rndsystems.com/cn/products/rat-igg-2a-isotype-control_mab006r

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	H146 cell line was obtained from American Type Culture Collection.
Authentication	The cell line has been authenticated by short tandem repeat profiling.
Mycoplasma contamination	The cell line has been validated to be free of mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	There is no ICLAC line used in this study.

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	PAD4flox/flox mice (B6(Cg)-Padi4tm1.2Kmw/J, 026708) and Ddx4-Cre mice (B6.FVB-Tg (Ddx4-cre)1Dcas/KnwJ, 018980) were purchased from the Jackson Laboratory. EPX-/- mice (C57BL/6-Epxem3(cre)Smoc/Smoc, NM-KO-190794) and MBP-/- (Prg2-/-) mice (C57BL/6-Prg2em1Smoc, NM-KO-190857) were purchased from the Shanghai Model Organisms Center, Inc (Shanghai, China). The CCDC25-/- mice with a C57BL/6 genetic background were generated by deleting the genomic DNA fragment covering exon 3 using a CRISPR-Cas9-mediated genome editing system by Shanghai Model Organisms Center. Two- to eight-week-old female wild-type C57BL/6 mice were purchased from the Animal Experiment Center of Sun Yat-Sen University. All mice were maintained at the Animal Experiment Center of Sun Yat-Sen University under the 14-hour lights on/10-hour lights off cycle and at 22°C and 60% humidity.
Wild animals	No wild animals were used.
Field-collected samples	BALF, peripheral blood, lung tissues or bone marrow cells were collected at the indicated time as shown in methods and figure legends. No field collected samples were used in the study.
Ethics oversight	All animal experiments were approved by the Institutional Review Boards and Animal Care and Use Committees of Sun Yat-Sen University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	Human BALF samples were obtained from 3 healthy volunteers including 1 male and 2 females with an average age of 27.7 years, and 46 asthma patients include 21 males and 25 females with an average age of 29.3 years from Sun Yat-Sen Memorial Hospital.
Recruitment	All samples were collected from healthy volunteers or patients with informed consent and all the participants received participant compensation. All the related procedures were performed with the approval of the internal review and ethics boards of Sun Yat-Sen Memorial Hospital. No self-selection bias in recruitment.
Ethics oversight	Internal review and ethics boards of Sun Yat-Sen Memorial Hospital.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

BALF cells were collected as described above. To obtain the total live cell number, BALF cells were resuspended in PBS and stained with 0.4% Trypan blue (Cat# 302643, Sigma-Aldrich; 1:2). After excluding dead cells labeled by trypan blue, the number of total live cells was counted using a hemacytometer (Cat# Z359629, Sigma-Aldrich). Non-specific antibody binding was blocked by addition of mouse Fc Block (Cat# 130-092-575, Miltenyi Biotec, 1:10) for 10 min at 4 °C. For detection of proportions of various immune cells, cells were stained with eBioscience™ Fixable Viability Dye eFluor™ 780 (Cat# 65-0865-14, ThermoFisher Scientific; 1:1000) in PBS for 30 min at 4 °C to eliminate dead cells. For surface staining, cells were incubated with FITC anti-CD3 (Cat# 100204, Biolegend; 1:20), PE anti-CD4 (Cat# 100408, Biolegend; 1:20), APC anti-SiglecF (Cat# 155508, Biolegend; 1:20), APC anti-CD11b (Cat# 101212, Biolegend, 1:20), APC anti-F4/80 (Cat# 123116, Biolegend, 1:20), FITC anti-Ly6G (Cat# 127606, Biolegend; 1:20), FITC anti-CD19 (Cat# 152403, Biolegend; 1:20), Pacific Blue anti-CD11c (Cat# 117322, Biolegend; 1:20) and FITC anti-MHC- II (Cat# 107606, Biolegend; 1:20) in PBS at 4°C for 30 min. For intracellular transcription factor staining, cells were fixed and permeabilized with eBioscience™ Transcription Factor Staining Buffer Set (Cat# 00-5523-00, ThermoFisher Scientific) according to the manufacturer's instruction and then stained with APC anti-GATA3 (Cat# 653806, Biolegend, 1:20) at 4°C for 30 min. For detection of EETs, cells were stained with 0.01 μM SYTOX Green (Cat# S7020, ThermoFisher Scientific) in Hank's Balanced Salt Solution for 15 min at room temperature in dark. After washing with PBS, cells were incubated with rabbit anti-histone H3 (citrulline 2,8,17) antibody (Cat# ab5103, Abcam; 1:30) and mouse anti-mouse ECP (Cat# MBS2090412, MyBioSource; 1:50), followed by immunofluorescent secondary antibody (Cat# A21428, A21235, ThermoFisher Scientific; 1:500) in PBS containing 1% BSA at 37°C for 1 hr respectively. Cells were washed with 2% BSA in PBS and centrifuged at 350 xg for 5 min at 4 °C after each incubation.

Instrument

CytoFLEX S

Software

FlowJo 10.0 (Treestar).

Cell population abundance

Percentages of F4/80+ cells, CD19+ cells, CD3+ cells, CD3+CD4+GATA3+ cells, CD11c-SiglecF+ cells, CD11b+Ly6G+ cells and F480-CD11c+MHC- II + cells in total live cells determined by flow cytometry were multiplied with the number of total live cells to quantify the number of macrophages, B cells, T cells, Th2, eosinophils, neutrophils and dendritic cells in BALF.

Gating strategy

The cells were gated on FSC-A/SSC-A and doublets were excluded using FSC-A/FSC-H. For quantitation of the various immune cells, the dead cells were eliminated by FVD. F4/80+ cells, CD19+ cells, CD3+ cells, CD3+CD4+GATA3+ cells, CD11c-SiglecF+ cells, CD11b+Ly6G+ cells and F480-CD11c+MHC- II + cells were gated. For detection of EETs, cells were gated on SYTOX Green +H3cit+ECP+.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.